General Relativity as a Dynamical PDE

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1 Introduction

Any time a new physical theory is proposed, a thorough analysis of the local as well as global existence and uniqueness of solutions to its equations of motion becomes necessary. In the case of general relativity, whose equations of motion are Einstein's equations, reading

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi T_{\mu\nu}, \tag{1.1}$$

this analysis becomes more involved, for multiple reasons.

The first and likely most obvious reason is that equation (1.1) is a nonlinear and highly coupled system of partial differential equations (PDEs) in the 10 component functions of the metric, the symmetric tensor field $g_{\mu\nu}$ (ignoring matter fields for now; this essay focuses on the vacuum case). Non-linear PDEs are typically more difficult to treat than their linear counterparts, often requiring a case-by-case analysis.

A further source of complexity can be observed in the following consideration. In, for example, the scalar wave equation

$$g^{\mu\nu} \nabla_\mu \nabla_\nu \phi = 0, \tag{1.2}$$

on some fixed Lorentzian background manifold $(\mathcal{M}, g)$, the (inverse) metric dictates the principal symbol of the equation, and by that the propagation of solutions ϕ within the background manifold. In a certain sense that we will explore in detail later on, Einstein's equations (1.1) are propagation equations for the tensor field $g_{\mu\nu}$, a priori making them appear similar to the wave equation. However, because it is still the metric which dictates the causal structure, general relativity is a theory of a tensor field which determines its own propagation. This remarkable feature of the theory has profound consequences for the global nature of solutions—the propagation of the metric actively shapes the spacetime manifold it is propagating in.

The fact that, fundamentally, general relativity is a geometric theory further complicates the situation. Concretely, this means that the theory is independent of coordinates. However, without a choice of coordinates, we cannot express general relativity as a PDE; the equation (1.1) is written with respect to a coordinate basis ∂_μ. Since, however, equation (1.1) is tensorial in nature, it is valid in any coordinate system. This makes the issue of solution uniqueness subtle, since the same initial data could have two different solutions related by a change of coordinates in a region away from the initial data slice. To be more precise, equation (1.1) has *gauge freedom* in the form of the four arbitrary coordinate functions picked on the spacetime manifold. This underdetermination of solutions might appear strange at first—after all, we have 10 equations determining the 10 unknown component functions of the metric. However, this apparent discrepancy is quickly resolved by recalling that the geometric identity

$$\nabla^\mu G_{\mu\nu} = \nabla^\mu (R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R) = 0 \tag{1.3}$$

as well as the condition of conservation on the energy-momentum tensor $\nabla^\mu T_{\mu\nu} = 0$. These impose four linear relations on the 10 equations in (1.1), explaining why solutions $g_{\mu\nu}$ are determined only up to a change of coordinates.

To alleviate this issue of underdetermination, one can follow an approach similar to that used in electromagnetism, which also has gauge freedom; we can fix the gauge using a gauge condition, which in the case of general relativity means imposing conditions on the coordinates. As we will see, a particularly useful gauge is ensured by the choice

$$\Box_g x^{\bar{\mu}} = 0. \tag{1.4}$$

This gauge, known as the *harmonic gauge*, brings the vacuum Einstein equations $R_{\mu\nu} = 0$ into a manifestly hyperbolic form, at which point standard PDE results can be employed to obtain local existence and uniqueness of solutions.

In this essay, we discuss how, starting from the vacuum Einstein equations,

$$R_{\mu\nu} = 0, \tag{1.5}$$

one of the most central results in mathematical relativity is established—the Choquet-Bruhat–Geroch theorem first published in [1] in 1969. Roughly speaking, this theorem states that provided valid initial data on some Cauchy surface Σ, there exists a unique maximal globally hyperbolic spacetime into which this surface embeds. Put differently, this result states that given initial data, there exists a unique largest spacetime that can be predicted from it. Uniqueness, here, is of course to be understood as up to isometry.

The structure of this essay is as follows. In Section 2, we outline the ADM formalism to obtain a geometric perspective allowing us to view general relativity as a dynamical PDE. In particular, this will allow us to distinguish gauge from dynamical degrees of freedom, the latter of which we identify to be the induced metric γ on the leaves of a foliation of $(\mathcal{M}, g)$ into spacelike hypersurfaces, as well as their associated extrinsic curvature K. In Section 3, we study how the requirement of $(\mathcal{M}, g)$ being a vacuum Einstein manifold translates to constraints on (γ, K), and show that if satisfied initially, they hold at all times, provided that the evolution equations hold. We then continue to introduce the harmonic gauge in Section 4, by means of a "guided rediscovery" of the gauge condition through the application of the so-called *Bochner formula*. Further, we consider how imposing the harmonic gauge condition promotes the a priori non-dynamical metric functions—the *lapse* α and the *shift* β^i, which are pure gauge—to dynamical variables, and how to relate initial data in the ADM picture to initial data in harmonic coordinates.

In Section 5, we work towards the Choquet-Bruhat–Geroch theorem presented in [1], establishing the existence and uniqueness of a maximal so-called *development* of initial data. After introducing the relevant definitions, we translate the local PDE-type existence and uniqueness results into geometric language, which provides us with the core input to the proof of the Choquet-Bruhat–Geroch theorem. Then follows said proof, split into three main parts; the first establishing the existence of a maximal development, and the second providing the existence and uniqueness of a maximal common subdevelopment of any two developments. This serves as an important tool in the proof of the third part, in which we show the uniqueness of the maximal development.

In the last section, we address how the Choquet-Bruhat–Geroch result [1] interacts with the strong cosmic censorship conjecture, as well as what kind of statement it makes (or rather, does not make) about the presence of singularities. Further, we consider how the employment of Zorn's lemma in the proof introduces the axiom of choice into the axiomatic system of general relativity, and point towards a more modern proof by Sbierski [2], which avoids Zorn's lemma by constructing maximal developments explicitly.

2 Geometry of the Evolution System

In this section, we introduce the ADM decomposition of the metric with the goal of identifying the dynamical quantities of general relativity. In addition, this decomposition provides us with a geometric viewpoint of the vacuum Einstein equations

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 0 \quad \Longleftrightarrow \quad R_{\mu\nu} = 0 \tag{2.1}$$

considered as a dynamical PDE system. To this end, we must pick a "time direction" along which we let its dynamics unfold; More concretely, we should pick a timelike coordinate t to evolve against. Ideally, this time coordinate is defined on the entire spacetime manifold $(\mathcal{M}, g)$—let us briefly review a sufficient condition for this to be possible.

Definition 2.1. *Let $(\mathcal{M}, g)$ be a spacetime manifold and $\Sigma \subseteq \mathcal{M}$ a hypersurface. Then,*

(i) Σ is called a Cauchy surface *if every causal curve in $(\mathcal{M}, g)$ intersects Σ exactly once;*

(ii) $(\mathcal{M}, g)$ is called globally hyperbolic *if it admits a Cauchy surface Σ.*

For globally hyperbolic spacetimes, the following important result holds:

Theorem 2.1. *(Geroch, Wald, Bernal & Sanchez) Let $(\mathcal{M}, g)$ be a globally hyperbolic spacetime. Then,*

(i) There exists a (non-unique) globally defined time function $t : \mathcal{M} \to \mathbb{R}$ such that the normal to surfaces of constant t, the vector $n = -(dt)^\sharp$, is timelike everywhere, and in particular, non-zero.

(ii) The surfaces of constant t, denoted by $\Sigma_t = \{p \in \mathcal{M} \mid t(p) = t\}$, are all Cauchy surfaces.

(iii) All surfaces Σ_t have the same topology, say Σ, such that $\mathcal{M} \cong \mathbb{R} \times \Sigma$.

This provides us with exactly what we need: a globally defined time coordinate, which *foliates* the spacetime $(\mathcal{M}, g)$ into the *leaves* or *slices* Σ_t. We note that $\{\Sigma_t\}_{t\in\mathbb{R}}$ is a family of non-intersecting spacelike hypersurfaces that cover $\mathcal{M}$, known as a *foliation*.

At this point, one might interject that global hyperbolicity is a rather strong assumption on the causal nature of the spacetime, one that we should not be able to make in the following. However, let us remind ourselves of what we are trying to achieve: we want to prove a result about the uniqueness of the maximal development of some initial data specified on a hypersurface Σ. To have uniqueness, we are limited to study (subsets of) the domain of dependence $D[\Sigma]$. Anywhere beyond it—on causal grounds—we cannot uniquely predict anything from data on Σ alone. By definition, however, Σ is a Cauchy surface of $D[\Sigma]$; this makes $(D[\Sigma], g)$ globally hyperbolic. In other words, for the context of studying general relativity as a dynamical PDE, we are safe to assume global hyperbolicity, and by that, are guaranteed to have a globally defined time coordinate t. Of course, larger, non-hyperbolic extensions of $(D[\Sigma], g)$ might have a certain kind of uniqueness—such as e.g. being the unique extension in the analytic class, where the requirement of analyticity fixes the a priori ambiguous choice of extension—but unique predictability from initial data, without such assumptions, can only be provided on the globally hyperbolic portion $(D[\Sigma], g)$.

2.1 ADM Decomposition of the Metric

The existence of this time coordinate t, or equivalently, the foliation $\{\Sigma_t\}$, is the starting point in the development of the *Arnowitt-Deser-Misner* or *ADM* formalism, first introduced in [3] and discussed in more detail in [4]. At the foundation of the formalism lies a geometric decomposition of the metric, which we now derive. We note that the normal 1-form of the foliation, dt, gives rise to a unique unit-length future-directed normal vector field,

$$n = -\alpha(dt)^\sharp. \tag{2.2}$$

The normalisation factor $\alpha > 0$ is known as the *lapse*, fixed by the relation

$$-1 = g(n, n) = \alpha^2 g^{tt} \quad \iff \quad g^{tt} = -\frac{1}{\alpha^2}. \tag{2.3}$$

Amending t to a local coordinate system (t, x^i) with x^i necessarily spacelike coordinates, we can expand the foliation-normal vector field as

$$n = -\alpha(dt)^\sharp = -\alpha g^{t\mu}\partial_\mu = -\alpha g^{tt}\partial_t - \alpha g^{ti}\partial_i = \frac{1}{\alpha}(\partial_t - \beta^i\partial_i) \tag{2.4}$$

where $\partial_i = \partial/\partial x^i$ are the coordinate basis vector fields tangent to the leaves Σ_t and the *shift* β^i was implicitly defined as $\beta^i = \alpha^2 g^{ti}$. Rearranging for ∂_t, we find

$$\partial_t = \alpha n + \beta, \quad \text{with} \quad \beta = \beta^i\partial_i. \tag{2.5}$$

Defining the components of the induced metric $\gamma = \iota^* g$ on the leaves as $\gamma_{ij} = g(\partial_i, \partial_j)$, we can evaluate the remaining metric components as

$$g_{tt} = g(\partial_t, \partial_t) = \alpha^2 \underbrace{g(n,n)}_{=-1} + 2\alpha \underbrace{g(n,\beta)}_{=0,\, n\perp\beta} + g(\beta,\beta) = -\alpha^2 + \gamma_{ij}\beta^i\beta^j, \tag{2.6}$$

$$g_{ti} = g(\partial_t, \partial_i) = \alpha g(n, \partial_i) + g(\beta, \partial_i) = \gamma_{ik}\beta^k. \tag{2.7}$$

We introduce the shorthand $\beta_i = \gamma_{ik}\beta^k$ for lowering spatial indices $i, j, k, \ldots$, which allows us to write the metric components in compact matrix form as

$$[g_{\mu\nu}] = \begin{pmatrix} -\alpha^2 + \beta_k\beta^k & \beta_j \\ \beta_i & \gamma_{ij} \end{pmatrix} \tag{2.8}$$

Alternatively, in tensorial form,

$$\begin{aligned} g &= (-\alpha^2 + \beta_k\beta^k)dt^2 + 2\beta_i dx^i dt + \gamma_{ij}dx^i dx^j \\ &= -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt) \end{aligned} \tag{2.9}$$

This is what is known as the *ADM decomposition of the metric* associated to the foliation $\{\Sigma_t\}$.

Occasionally, it is useful to have the components of the inverse metric on hand. A straightforward calculation reveals that

$$[g^{\mu\nu}] = \begin{pmatrix} -1/\alpha^2 & \beta^j/\alpha^2 \\ \beta^i/\alpha^2 & \gamma^{ij} - \beta^i\beta^j/\alpha^2 \end{pmatrix} \tag{2.10}$$

where γ^{ij} are the components of the 3×3-inverse of γ_{ij} subject to

$$\gamma^{ik}\gamma_{kj} = \delta^i{}_j. \tag{2.11}$$

Alternatively, we have the basis expansion

$$\begin{aligned} g^{-1} &= -\frac{1}{\alpha^2}(\partial_t - \beta) \otimes (\partial_t - \beta) + \gamma^{ij}\partial_i \otimes \partial_j \\ &= -n \otimes n + \gamma^{ij}\partial_i \otimes \partial_j \end{aligned} \tag{2.12}$$

Another useful quantity is the metric determinant, which reads $\sqrt{g} = \alpha\sqrt{\gamma}$ with $\sqrt{\gamma} = \sqrt{\det \gamma_{ij}}$.

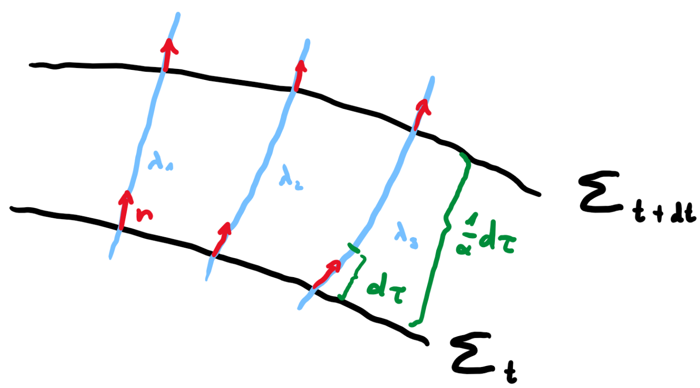
2.2 Geometric Interpretation and Identification of Dynamical Variables

The ADM decomposition (2.9) of the metric has recast its components in terms of the lapse α, the shift β and the induced spatial metric γ. This is particularly useful because these three quantities may be interpreted geometrically, allowing us to distinguish the dynamical from the pure gauge variables:

- **Lapse** α: Consider an observer traveling along an integral line $\lambda(u) : I \to \mathcal{M}$ of the foliation-normal vector field n, that is, a solution to

$$\dot{\lambda}(u) := \frac{d}{du}\lambda(u) = n(\lambda(u)). \tag{2.13}$$

Since n is unit-normalised, we note that $g(\dot{\lambda}, \dot{\lambda}) = g(n, n) = -1$, implying that $u = \tau$ is proper time. We may sketch this as follows:



(2.14)

Here, the blue lines indicate the integral lines λ of the foliation-normal vector field n.

We further observe that the 1-form $\beta^i dt + dx^i$ annihilates n, since

$$(\beta^i dt + dx^i)(n) = \beta^i dt(\alpha^{-1}\partial_t) + dx^i(-\alpha^{-1}\beta^k\partial_k) = \alpha^{-1}(\beta^i - \beta^k\delta^i{}_k) = 0 \tag{2.15}$$

Consequently, the metric pulls back to $g|_\lambda = -\alpha^2 dt^2$ on λ, implying

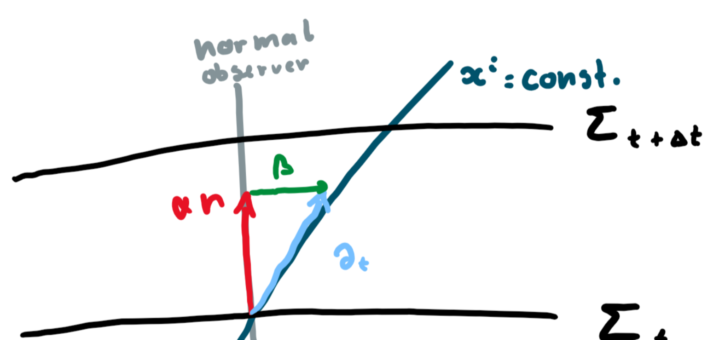
$$d\tau^2 = \alpha^2 dt^2 \quad \Longleftrightarrow \quad d\tau = \pm\alpha dt. \tag{2.16}$$

This means that α quantifies the ratio between the passage of coordinate time t and proper time τ for so-called *Eulerian observers* traveling along the normal flow of the foliation. As such, it can be changed at will by redefinition of the t-coordinate, and is hence a pure gauge quantity.

- **Shift** β: We recall from the above that

$$\partial_t = \alpha n + \beta. \tag{2.17}$$

This splits the coordinate basis vector ∂_t into a normal and a tangential component. Hence, the shift β measures the failure of ∂_t to be normal to the foliation. As such, it quantifies the rate of divergence between integral curves of ∂_t and αn:



(2.18)

A characterising property of the integral curves of the vector field ∂_t is that they are the coordinate lines of t, i.e. the curves obtained by fixing a value of the spatial coordinates x^i and varying t. That is, the graph of such a curve λ in $\mathcal{M}$ takes the form

$$\Gamma(\lambda) = \{p \in \mathcal{M} \mid x^i(p) = x^i_0\} \tag{2.19}$$

for some constant choice of x^i_0. Concretely, this means that the integral curves of ∂_t are determined (up to reparametrisation) by the coordinate functions $x^i(p)$.

Thus, β measures the rate of *shift* of the spatial coordinates x^i relative to the normal flow generated by αn. By the above argument, this shift is dependent on the choice of spatial coordinates—it is *pure gauge*. In particular, as is shown in [5], one can bring the metric into the form

$$g = -dt^2 + \gamma_{ij} dx^i dx^j \tag{2.20}$$

by means of constructing so-called *Gaussian normal coordinates*. This is the ADM-decomposed form of the metric (2.9) with the specific values of $\alpha = 1$ and $\beta = 0$, further reinforcing that these quantities are pure gauge.

- **Spatial Metric** γ: The third ADM quantity, γ, is the pullback of the ambient metric g onto the time-slices Σ_t. It therefore contains geometric information about g, specifically through the defining property of the pullback that

$$\gamma(X, Y) = g(X, Y) \tag{2.21}$$

for all foliation-tangent vector fields $X, Y \in \Gamma(T\Sigma)$. Once the foliation $\{\Sigma_t\}$ is fixed, $\gamma|_{\Sigma_t}$ is a Riemannian metric on the 3-manifold Σ_t, which inherits its geometric structure from the ambient manifold $(\mathcal{M}, g)$. Though the *components* γ_{ij} depend on the choice of coordinates (x^i), the geometric information pulled back from the ambient spacetime does not—it is generally impossible to bring them into a standard form like $\gamma_{ij} = \delta_{ij}$. This allows us to identify γ as the dynamical variable of the theory.

In a second-order system like general relativity, it is common to consider the time-derivative of the dynamical variable as an independent dynamical degree of freedom—in this case, $\partial_t \gamma_{ij}$—to obtain a first-order system and to facilitate prescription of initial data. For ease of geometric interpretation, however, it is useful to "repackage" this time-derivative information of γ into a tensorial quantity; unlike γ_{ij}, the functions $\partial_t \gamma_{ij}$ are—generally speaking—not the components of a rank-2 tensor on the foliation. Further, treating $\partial_t \gamma_{ij}$ as a dynamical variable is geometrically opaque—it is unclear what its interpretation is.

To alleviate these issues, we introduce the *extrinsic curvature* of the foliation by

$$K = -\frac{1}{2} \iota^*(\mathcal{L}_n g), \tag{2.22}$$

where $\mathcal{L}_n$ denotes the Lie-derivative along the foliation-normal vector field n, and ι^* is the pullback map onto the foliation's leaves Σ_t. Clearly, K is a 2-tensor on the tangent bundle $T\Sigma$ of the foliation. Before giving it a geometric interpretation, let us verify that it actually contains the time-derivative information $\partial_t \gamma_{ij}$ in a useful way. To this end, we note that in the coordinates (t, x^i), the pullback ι^* essentially consists of ignoring any t-components of K, leaving only

$$K_{ij} = -\frac{1}{2} (\mathcal{L}_n g)_{ij} = -\frac{1}{2} \big(n[\gamma_{ij}] + (\partial_i n^\mu) g_{\mu j} + (\partial_j n^\mu) g_{i\mu} \big). \tag{2.23}$$

To simplify this, we expand the second term as

$$
\begin{aligned}
(\partial_i n^\mu) g_{\mu j} &= (\partial_i n^t) g_{tj} + (\partial_i n^k)\gamma_{kj} = \partial_i \left(\frac{1}{\alpha}\right)\beta_j - \partial_i\left(\frac{\beta^k}{\alpha}\right)\gamma_{kj} \\
&= \cancel{\partial_i \left(\frac{1}{\alpha}\right)\beta_j} - \cancel{\partial_i \left(\frac{1}{\alpha}\right)\beta^k\gamma_{kj}} - \frac{1}{\alpha}(\partial_i\beta^k)\gamma_{kj},
\end{aligned}
\tag{2.24}
$$

with the third term expanding analogously. Reinserting back into equation (2.23), we obtain

$$
\begin{aligned}
K_{ij} &= -\frac{1}{2\alpha}\big(\partial_t\gamma_{ij} - \beta^k\partial_k\gamma_{ij} - (\partial_i\beta^k)\gamma_{kj} - (\partial_j\beta^k)\gamma_{ik}\big) \\
&= -\frac{1}{2\alpha}\big(\partial_t\gamma_{ij} - (\bar{\mathcal{L}}_\beta\gamma)_{ij}\big) = -\frac{1}{2\alpha}\big(\partial_t\gamma_{ij} - \bar\nabla_i\beta_j - \bar\nabla_j\beta_i\big)
\end{aligned}
\tag{2.25}
$$

Here, $\bar{\mathcal{L}}_\beta$ denotes the Lie derivative along β and $\bar\nabla$ the induced connection, both intrinsic to the leaves of the foliation. Clearly, once α, β are fixed by a choice of gauge, knowing the components (γ_{ij}, K_{ij}) is equivalent to knowing the values of $(\gamma_{ij}, \partial_t\gamma_{ij})$, but K_{ij} has the advantage of being a tensorial quantity.

Having verified that K does indeed contain first-order time-derivative information about γ, let us now turn to its geometric interpretation. Coordinate-free notation lends itself particularly well for this task—for foliation-tangent vector fields $X, Y \in \Gamma(T\Sigma)$, we have

$$
\begin{aligned}
K(X,Y) &= -\frac{1}{2}(\mathcal{L}_n g)(X,Y) = -\frac{1}{2}\Big(\nabla_n(g(X,Y)) - g([n,X],Y) - g(X,[n,Y])\Big) \\
&= -\frac{1}{2}\Big(\cancel{g(\nabla_n X, Y)} - g(\cancel{\nabla_n X} - \nabla_X n, Y)\Big) + (X \leftrightarrow Y) \\
&= \frac{1}{2}g(n, \nabla_X Y + \nabla_Y X) = \frac{1}{2}g(n, 2\nabla_X Y + [X,Y]).
\end{aligned}
\tag{2.26}
$$

Making use of the fact that $[X,Y] \in \Gamma(T\Sigma)$ and integrating by parts once more, we find the identity

$$
K(X,Y) = -g(\nabla_X n, Y). \tag{2.27}
$$

Thus, $K(X,Y)$ measures the rate of change of the normal vector field n when moving tangentially to a leaf along X, projected onto the direction Y. As such, K encodes how the leaves Σ_t "bend" within the ambient space $(\mathcal{M}, g)$ they are embedded in. Thus, roughly speaking, K tells us how a leaf Σ_t is "shaped" within the ambient spacetime, whereas γ describes the geometry intrinsic to it. With this interpretation, it makes intuitive sense that (γ, K) are the dynamical variables: merely knowing the geometry of a hypersurface is not sufficient—we also have to know how the hypersurface is shaped as it is embedded within the ambient space.

This discussion further shows that prescribing specific values for (γ, K) on a leaf, say $\Sigma_{t=0}$, is a geometrically natural way of imposing initial data. More specifically, given a Riemannian 3-manifold (Σ, γ) and a symmetric $(0,2)$-tensor field K on Σ, we say that $(\mathcal{M}, g)$ is a development of the initial data (Σ, γ, K) if there exists an isometric embedding $\iota : \Sigma \to \mathcal{M}$ whose extrinsic curvature in $(\mathcal{M}, g)$ pulls back to K on Σ. We will return to make this more precise later.

3 Constraints

We now turn to an important aspect of general relativity—the fact that it is a *constrained* system. In essence, what this means is that not just any configuration of the dynamical variables (γ, K) is physically allowed—they must satisfy so-called *constraint equations*. In this section, we discuss how they arise geometrically, derive them explicitly, and show that if initial data satisfies them, they are preserved by evolution under the Einstein equations.

3.1 Derivation of the Constraint Equations

Already on intuitive grounds, it becomes apparent that the ambient Riemann curvature associated to $(\mathcal{M}, g)$, the intrinsic Riemann curvature associated to (Σ_t, γ), as well as the extrinsic curvature K, must be related in some way. One can convince oneself of this through the following argument: If the ambient manifold $(\mathcal{M}, g)$ curves "under" a hypersurface Σ, the induced geometry on Σ must experience some of this curvature as well, providing a (so far only qualitative) link between the ambient and intrinsic curvatures. From examples such as $S^2 \subseteq \mathbb{R}^3$, however, we know that it is possible for a curved manifold, S^2, to live on a flat background, $\mathbb{R}^3$. Here, the curvature comes from how S^2 itself *curves* as it is embedded into $\mathbb{R}^3$—something we identified the extrinsic curvature to describe.

Concretely, these relationships are described by the *Gauss-Codazzi-Mainardi* equations. These are derived and discussed in detail in e.g. [6]—for the purposes of this essay, we will only state the first two of these three equations, as this is all we need to discuss constraints. In coordinate-free language, these equations read

$$R(X,Y,Z,W) = \bar{R}(X,Y,Z,W) + K(X,Z)K(Y,W) - K(X,W)K(Y,Z), \tag{3.1}$$

$$R(n,Z,X,Y) = (\bar{\nabla}_X K)(Y,Z) - (\bar{\nabla}_Y K)(X,Z), \tag{3.2}$$

with the former known as the *Gauss equation* and the latter as the *Codazzi equation*. Here, R denotes the *ambient* Riemann tensor associated to the Levi-Civita connection on $(\mathcal{M}, g)$, $\bar{\nabla}$ the induced connection on the leaves which is the Levi-Civita connection on (Σ_t, γ), and $\bar{R}$ its associated *intrinsic* curvature.

In what follows, it will be useful to work in a basis (n, ∂_i), denoting an n-index by $\perp$ and spatial indices by $i, j, k, \dots$. In this basis, the Gauss–Codazzi equations read

$$R_{ijk\ell} = \bar{R}_{ijk\ell} + K_{ik}K_{j\ell} - K_{i\ell}K_{jk}, \tag{3.3}$$

$$R_{\perp kij} = \bar{\nabla}_i K_{jk} - \bar{\nabla}_j K_{ik}. \tag{3.4}$$

To derive the constraints from Einstein's equations, we will additionally need the metric components in this basis to form contractions. From equation (2.12), it is easy to see that the inverse metric takes the block-diagonal form

$$g^{\perp\perp} = -1, \quad g^{i\perp} = 0, \quad g^{ij} = \gamma^{ij}, \tag{3.5}$$

whence the metric has components

$$g_{\perp\perp} = -1, \quad g_{i\perp} = 0, \quad g_{ij} = \gamma_{ij}. \tag{3.6}$$

This allows us to relate the ambient Einstein tensor components to the intrinsic and extrinsic curvature as follows:

$$\begin{aligned} G_{\perp\perp} &= R_{\perp\perp} - \frac{1}{2} g_{\perp\perp} R = g^{\mu\nu} R_{\mu\perp\nu\perp} + \frac{1}{2} g^{\mu\nu} g^{\rho\sigma} R_{\mu\rho\nu\sigma} = \cancel{\gamma^{ij} R_{i\perp j\perp}} + \frac{1}{2}\Big(\gamma^{ik}\gamma^{j\ell} R_{ijk\ell} - \cancel{2\gamma^{ij} R_{i\perp j\perp}}\Big) \\ &= \frac{1}{2}\gamma^{ik}\gamma^{j\ell}\Big(\bar{R}_{ijk\ell} + K_{ik}K_{j\ell} - K_{i\ell}K_{jk}\Big) = \frac{1}{2}\big(\bar{R} + K^2 - K_{ij}K^{ij}\big), \end{aligned} \tag{3.7}$$

and

$$\begin{aligned} G_{\perp i} &= R_{\perp i} - \frac{1}{2} \underbrace{g_{\perp i}}_{=0} R = g^{\mu\nu} R_{\mu\perp\nu i} = g^{\perp\perp} \underbrace{R_{\perp\perp\perp i}}_{=0} + \gamma^{kj} R_{k\perp ji} = \gamma^{jk}(\bar{\nabla}_i K_{jk} - \bar{\nabla}_j K_{ik}) \\ &= \bar{\nabla}_i K - \bar{\nabla}^k K_{ik} \end{aligned} \tag{3.8}$$

Here, $\bar{R}$ is the intrinsic Ricci scalar and $K = \gamma^{ij} K_{ij}$ the mean curvature of the foliation.

For a vacuum solution, the Einstein equations prescribe $G_{\mu\nu} = 0$, so in particular, $G_{\perp\perp} = G_{\perp i} = 0$. Hence, the dynamical degrees of freedom (γ, K) are not allowed to have arbitrary configurations—they must satisfy

$$\bar{R} + K^2 - K_{ij}K^{ij} = 0, \qquad \bar{\nabla}_i(K^{ij} - \gamma^{ij}K) = 0. \tag{3.9}$$

Noteworthily, when considering γ_{ij} and K_{ij} as independent degrees of freedom, these constraints do not contain any time derivatives. Instead of encoding evolution, they must be satisfied at all times *alongside* the evolution equations, which—by exclusion—are the remaining six equations $G_{ij} = 0$. For this reason, the above equalities are known as the *Hamiltonian* and *momentum constraints*, respectively. Intuitively speaking, they ensure that (Σ, γ, K) curves in such a way that it is possible to embed it into a vacuum spacetime $(\mathcal{M}, g)$.

Remarks:

- General relativity's structure as a constrained system is akin to electrodynamics: the full set of source-free Maxwell's equations read (in appropriate units)

  $$\begin{aligned} \nabla \cdot \mathbf{E} &= 0, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\partial_t \mathbf{B}, & \nabla \times \mathbf{B} &= \partial_t \mathbf{E}. \end{aligned} \tag{3.10}$$

  The two equations on the right are satisfied automatically when writing $(\mathbf{E}, \mathbf{B})$ in terms of the standard potentials $(\phi, \mathbf{A})$, leaving a single constraint equation, $\nabla \cdot \mathbf{E} = 0$, and three evolution equations, $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$. This is compared to four constraints and six evolution equations in general relativity.

- For a solution of the sourced Einstein equations $G_{\mu\nu} = 8\pi T_{\mu\nu}$, the constraint equations generalise to

  $$\bar{R} + K^2 - K_{ij}K^{ij} = 16\pi\rho, \qquad \bar{\nabla}_i(K^{ik} - \gamma^{ik}K) = -8\pi j^k, \tag{3.11}$$

  where $\rho = T_{\perp\perp} = T_{\mu\nu}n^\mu n^\nu$ and $j^k = T^k{}_\perp = -T^k{}_\mu n^\mu$ are the energy and momentum densities as measured by Eulerian observers traveling along the integral lines of the foliation-normal vector field n.

- Using the Gauss-Codazzi equations, one can further show that the Einstein-Hilbert action takes the form

  $$S_{\mathrm{EH}}[\alpha, \beta, \gamma] = \int_{\mathcal{M}} d^4x\, \alpha\sqrt{\gamma}\Big(\bar{R} + K_{ij}K^{ij} - K^2\Big) + (\text{boundary terms}) \tag{3.12}$$

  We note that only γ appears with time derivatives through the presence of the extrinsic curvature terms; the lapse α and shift β only appear with spatial derivatives, making them non-dynamical. This further confirms what we have argued for on geometric grounds in Section 2.2.

3.2 Consistency of the Constraints

Since the Einstein equations must hold at all times, it follows that the constraints must be satisfied on each time slice Σ_t. In particular, as such, the constraints must hold for the initial data. For general relativity to make sense as an evolution system, the constraints should propagate; if they hold initially, they should continue to hold under time evolution. In other words, we expect the constraints to be *consistent*.

Alcubierre [4] gives an elegant way of showing that this is, in fact, the case. In this section, we present this short derivation in an adapted form. We start by noting that the Bianchi identities $\nabla_\mu G^\mu{}_\nu = 0$ imply that

$$\nabla_\perp G_{\perp j} = -\nabla_\perp G^\perp{}_j = \nabla_i G^i{}_j, \quad \nabla_\perp G_{\perp\perp} = -\nabla_\perp G^\perp{}_\perp = \nabla_i G^i{}_\perp \tag{3.13}$$

Assuming that the evolution equations $G_{ij} = 0$ hold, we can make explicit the right-hand side of the first equation

$$\begin{aligned}\nabla_i G^i{}_j &= \underbrace{\partial_i G^i{}_j}_{=0} + \Gamma^i{}_{\perp i} G^\perp{}_j + \Gamma^i{}_{ki} \underbrace{G^k{}_j}_{=0} - \Gamma^\perp{}_{ji} G^i{}_\perp - \Gamma^k{}_{ji} \underbrace{G^i{}_k}_{=0} \\ &= \Gamma^i{}_{\perp i} G^\perp{}_i - \Gamma^\perp{}_{ji} G^i{}_\perp\end{aligned} \tag{3.14}$$

For the left-hand side, we obtain

$$\nabla_\perp G_{\perp j} = n^\mu \partial_\mu G_{\perp j} - \Gamma^\perp{}_{\perp\perp} G_{\perp j} - \Gamma^\perp{}_{j\perp} G_{\perp\perp} - \Gamma^k{}_{j\perp} G_{\perp k}. \tag{3.15}$$

An analogous treatment for the second equation reveals that given $G_{ij} = 0$, the Bianchi identities imply a first-order linear evolution system for the constraints, which can be written schematically as

$$\begin{aligned}n^\mu \partial_\mu G_{\perp j} &= \Gamma \cdot G_{\perp j} + \Gamma \cdot G_{\perp\perp}, \\ n^\mu \partial_\mu G_{\perp\perp} &= \partial_i G^i{}_\perp + \Gamma \cdot G_{\perp j} + \Gamma \cdot G_{\perp\perp}\end{aligned} \tag{3.16}$$

Specifically, if the constraints are satisfied on some initial time slice Σ, i.e. $G_{\perp j} = G_{\perp\perp} = 0$ initially, then according to this linear system (3.16), they remain satisfied along the normal evolution of the foliation. Since the foliation-normal flow moves between leaves, this means that if the constraints are satisfied on one leaf of the foliation, they are satisfied on any.

Although from a mathematical point of view, the consistency of the constraints is a "done deal", we should remark that in numerical relativity, it is unfortunately not that simple. This is because when we work with the initial data (Σ, γ, K) on a computer, we can only solve the constraints up to machine precision at best—we only have $G_{\perp i} \approx 0$, $G_{\perp\perp} \approx 0$. The so-called *constraint-violating modes* propagate as well, potentially growing in magnitude; further, the floating-point errors in each iteration step might introduce additional constraint-violating modes. This poses an issue which needs to be addressed for long-term numerical evolutions of gravitational systems, which we will not go into any detail on in the following.

4 Harmonic Gauge

As written in their most general form, the vacuum Einstein equations

$$R_{\mu\nu} = 0 \tag{4.1}$$

do not have any distinctive character, making them difficult to approach for proving existence and uniqueness results. Ideally, we would want something hyperbolic; From a mathematical perspective, hyperbolic equations are reasonably well-understood, and since we expect general relativity to describe the propagation of radiation, this would also make sense on physical grounds.

One might think that the Einstein equations could be brought into hyperbolic form by computing the equations of motion from the ADM form of the Einstein-Hilbert action (3.12) we briefly discussed before, upon choosing a "convenient" gauge such as $\alpha = 1$, $\beta = 0$. Unfortunately, however, as is discussed in chapter 5 of [4], the resulting ADM equations are only weakly hyperbolic and not well-posed. Although there exist well-posed modifications of ADM, such as the BSSN formulation, these are more complex and, to some degree, "optimised" for numerical relativity. In the following, rather, we are concerned with much more theoretical local as well as global existence and uniqueness results. We thus take a step back from ADM and seek to find a gauge in which $R_{\mu\nu} = 0$ takes a desirable form to which we can apply well-known PDE theorems. We achieve this by picking the so-called *harmonic gauge*, which turns the vacuum Einstein equations into a second-order quasilinear hyperbolic PDE. As an aside, we will then briefly return to the ADM formulation of general relativity and see how imposing the harmonic gauge promotes the lapse α and the shift β to *dynamical* variables.

4.1 Bochner Formula and the Ricci Tensor

Technically, one can approach the issue of turning equation (4.1) into a manifestly hyperbolic form by manually rearranging and manipulating terms in the full Ricci tensor expression

$$R_{\mu\nu} = \partial_\lambda \Gamma^\lambda{}_{\mu\nu} - \partial_\nu \Gamma^\lambda{}_{\mu\lambda} + \Gamma^\lambda{}_{\rho\lambda}\Gamma^\rho{}_{\mu\nu} - \Gamma^\lambda{}_{\rho\mu}\Gamma^\rho{}_{\lambda\nu}. \tag{4.2}$$

This is, however, cumbersome and not very elegant. In this section, inspired by [7], we entertain a different approach, in which the harmonic gauge will appear naturally as a favourable gauge choice.

The central result used for this approach is the so-called *Bochner formula* which we state and prove below:

Lemma 4.1. *(Bochner's Formula) For two scalar fields $\varphi, \psi \in C^\infty(\mathcal{M})$, it holds that*

$$\begin{aligned}\frac{1}{2}\Box_g(\nabla_a\varphi\nabla^a\psi) = \frac{1}{2}(\nabla_a\Box_g\varphi)\nabla^a\psi + \frac{1}{2}\nabla_a\varphi(\nabla^a\Box_g\psi)\\ + \nabla_a\nabla_b\varphi\nabla^a\nabla^b\psi + R^{ab}\nabla_a\varphi\nabla_b\psi.\end{aligned} \tag{4.3}$$

Here, $\Box_g = g^{ab}\nabla_a\nabla_b$ denotes the wave operator/Laplacian on the spacetime $(\mathcal{M}, g)$.

Before proving this result, let us briefly motivate why it is useful to us. Recall that our goal is to find a convenient expression for the Ricci tensor. Bochner's formula contains a term involving R^{ab}, which we may isolate by picking the scalars to be coordinate functions, $\varphi = x^{\bar{\mu}}$, $\psi = x^{\bar{\nu}}$, whence the term reduces to $R^{\bar{\mu}\bar{\nu}}$ (we will address this in more detail shortly). Through this choice, Bochner's formula provides us with a relationship between (differential operators acting on) the coordinate functions and the Ricci tensor components in these coordinates, which we can solve for $R^{\bar{\mu}\bar{\nu}}$.

Proof. We start with the left-hand side of equation (4.3) and expand in terms of covariant derivatives and the metric to obtain

$$\begin{aligned}\frac{1}{2}\Box_g(\nabla_a\varphi\nabla^a\psi) &= \frac{1}{2}g^{ab}g^{cd}\nabla_a\nabla_b(\nabla_c\varphi\nabla_d\psi) = \frac{1}{2}g^{ab}g^{cd}\nabla_a(\nabla_b\nabla_c\varphi\nabla_d\psi + \nabla_c\varphi\nabla_b\nabla_d\psi)\\ &= \frac{1}{2}g^{ab}g^{cd}(\nabla_a\nabla_b\nabla_c\varphi\nabla_d\psi + \nabla_b\nabla_c\varphi\nabla_a\nabla_d\psi) + (\varphi \leftrightarrow \psi)\end{aligned} \tag{4.4}$$

The second term in the parentheses, once symmetrised in φ, ψ, clearly contributes the $\nabla_a\nabla_b\varphi\nabla^a\nabla^b\psi$ term in the final result. Let us study the first term in more detail. Concretely, making use of

$$\nabla_a\nabla_b\varphi = \nabla_b\nabla_a\varphi, \quad [\nabla_a, \nabla_b]X_c = -R^d{}_{cab}X_d, \tag{4.5}$$

we rewrite

$$\begin{aligned}g^{ab}g^{cd}\nabla_a\nabla_b\nabla_c\varphi\nabla_d\psi &= g^{ab}g^{cd}\nabla_a\nabla_c\nabla_b\varphi\nabla_d\psi\\ &= g^{ab}g^{cd}(\nabla_c\nabla_a\nabla_b\varphi\nabla_d\psi + \underbrace{[\nabla_a, \nabla_c]\nabla_b\varphi}_{=-R^e{}_{bac}\nabla_e\varphi}\nabla_d\psi)\\ &= (\nabla_a\Box_g\varphi)\nabla^a\psi + R^{ab}\nabla_a\varphi\nabla_b\psi.\end{aligned} \tag{4.6}$$

Reinserting this into equation (4.4), we conclude that

$$\begin{aligned}\frac{1}{2}\Box_g(\nabla_a\varphi\nabla^a\psi) = \frac{1}{2}(\nabla_a\Box_g\varphi)\nabla^a\psi + \frac{1}{2}\nabla_a\varphi(\nabla^a\Box_g\psi)\\ + \nabla_a\nabla_b\varphi\nabla^a\nabla^b\psi + R^{ab}\nabla_a\varphi\nabla_b\psi.\end{aligned} \tag{4.7}$$

$\Box$

To isolate the Ricci tensor components, as outlined before, we make the choice

$$\varphi = x^{\bar{\mu}}, \quad \psi = x^{\bar{\nu}}, \tag{4.8}$$

that is, we let φ, ψ be coordinate functions. The bar on the indices is put in place to indicate that they are not to be interpreted as contravariant but merely as a label for the specific coordinate we are considering. This has a few crucial consequences. Firstly, for first-order covariant derivatives of the coordinate functions, we have

$$\nabla_\mu x^{\bar{\nu}} = \partial_\mu x^{\bar{\nu}} = \delta^{\bar{\nu}}{}_\mu, \tag{4.9}$$

since we are acting on the scalar $x^{\bar{\nu}}$ rather than interpreting it as the component of a vector. Taking a second covariant derivative, however, connection coefficients emerge, since the first derivative introduced a covariant index:

$$\nabla_\mu \nabla_\nu x^{\bar{\lambda}} = \nabla_\mu \delta^{\bar{\lambda}}{}_\nu = \underbrace{\partial_\mu \delta^{\bar{\lambda}}{}_\nu}_{=0} - \Gamma^\rho{}_{\nu\mu} \delta^{\bar{\lambda}}{}_\rho = -\Gamma^{\bar{\lambda}}{}_{\nu\mu} \tag{4.10}$$

We now insert our choice for φ, ψ into the Bochner formula (4.3) and rearrange for $R^{\bar{\mu}\bar{\nu}}$, yielding

$$\begin{aligned} R^{\bar{\mu}\bar{\nu}} &= R^{\rho\sigma} \delta^{\bar{\mu}}{}_\rho \delta^{\bar{\nu}}{}_\sigma = R^{\rho\sigma} \nabla_\rho x^{\bar{\mu}} \nabla_\sigma x^{\bar{\nu}} \\ &= \frac{1}{2} \Box_g (g^{\rho\sigma} \nabla_\rho x^{\bar{\mu}} \nabla_\sigma x^{\bar{\nu}}) - \nabla_\rho \nabla_\sigma x^{\bar{\mu}} \nabla^\rho \nabla^\sigma x^{\bar{\nu}} - \frac{1}{2} \big((\nabla_\lambda \Box_g x^{\bar{\mu}}) \nabla^\lambda x^{\bar{\nu}} + \nabla^\lambda x^{\bar{\mu}} (\nabla_\lambda \Box_g x^{\bar{\nu}}) \big) \end{aligned} \tag{4.11}$$

At this point, we can let the unwieldy nature of this expression motivate us to make a gauge choice: the last term vanishes entirely if we choose to require the coordinates to be *harmonic*, that is,

$$\Box_g x^{\bar{\mu}} = 0. \tag{4.12}$$

This is known as the *harmonic gauge.* Let us continue to simplify the expression for $R^{\bar{\mu}\bar{\nu}}$ using equations (4.9) and (4.10) as

$$R^{\bar{\mu}\bar{\nu}} = \frac{1}{2} \Box_g (g^{\rho\sigma} \delta^{\bar{\mu}}{}_\rho \delta^{\bar{\nu}}{}_\sigma) - (-\Gamma^{\bar{\mu}}{}_{\rho\sigma})(-\Gamma^{\bar{\nu}\rho\sigma}) = \frac{1}{2} \Box_g g^{\bar{\mu}\bar{\nu}} - \Gamma^{\bar{\mu}}{}_{\rho\sigma} \Gamma^{\bar{\nu}\rho\sigma} \tag{4.13}$$

At this point, one might think that the first term vanishes, since $\nabla g = 0$ for the Levi-Civita connection. This, however, is false: Both of the indices have a bar, whence we are acting on the inverse metric's component *function* $g^{\bar{\mu}\bar{\nu}}$, *treated as a scalar.* Explicitly, this evaluates to

$$\Box_g g^{\bar{\mu}\bar{\nu}} = g^{\rho\sigma} \nabla_\rho \nabla_\sigma g^{\bar{\mu}\bar{\nu}} = g^{\rho\sigma} (g^{\bar{\mu}\bar{\nu}}{}_{,\rho\sigma} - \Gamma^\lambda{}_{\rho\sigma} g^{\bar{\mu}\bar{\nu}}{}_{,\lambda}) = g^{\rho\sigma} g^{\bar{\mu}\bar{\nu}}{}_{,\rho\sigma} \tag{4.14}$$

In the last equality, we were able to drop the first-order term in $g^{\bar{\mu}\bar{\nu}}$—it vanishes due to

$$0 = \Box_g x^{\bar{\mu}} = g^{\rho\sigma} \nabla_\rho \nabla_\sigma x^{\bar{\mu}} = -g^{\rho\sigma} \Gamma^{\bar{\mu}}{}_{\rho\sigma}. \tag{4.15}$$

Consequently, the contravariant Ricci tensor in harmonic gauge reads

$$R^{\mu\nu} = \frac{1}{2} g^{\rho\sigma} g^{\mu\nu}{}_{,\rho\sigma} - \Gamma^\mu{}_{\rho\sigma} \Gamma^{\nu\rho\sigma}, \tag{4.16}$$

where we were able to omit the bar notation for indices since only partial derivatives remain at this stage.

Although equation (4.16) already gives us some insight towards the fact that in harmonic gauge, due to the Lorentzian signature of the (inverse) metric, the vacuum Einstein equations are of hyperbolic nature, its form is not as transparent as it can be. The issue is simple, but needs to be addressed; the second derivatives appearing in the first term are acting on the *inverse* metric rather than the metric itself. However, the quantity of interest in general relativity—though intimately related to its inverse—is the metric. We should hence rewrite equation (4.16) in terms of second derivatives of the metric. As a first step, we lower the indices, obtaining

$$R_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} g_{\mu\alpha} g_{\nu\beta} g^{\alpha\beta}{}_{,\rho\sigma} - \Gamma_{\mu\rho\sigma} \Gamma_{\nu}{}^{\rho\sigma} \tag{4.17}$$

To simplify the first term further, we recall that

$$\partial_\gamma g^{\alpha\beta} = -g^{\alpha\delta} g^{\beta\epsilon} \partial_\gamma g_{\delta\epsilon}, \tag{4.18}$$

whence the second-order term in the Ricci tensor above turns into

$$\begin{aligned}
\frac{1}{2} g^{\lambda\rho} g_{\mu\alpha} g_{\nu\beta} \partial_\lambda \partial_\rho g^{\alpha\beta} &= -\frac{1}{2} g^{\lambda\rho} g_{\mu\alpha} g_{\nu\beta} \partial_\lambda (g^{\alpha\delta} g^{\beta\epsilon} \partial_\rho g_{\delta\epsilon}) \\
&= -\frac{1}{2} g^{\lambda\rho} \partial_\lambda \partial_\rho g_{\mu\nu} - \frac{1}{2} g^{\lambda\rho} \left[(g_{\mu\sigma} \partial_\lambda g^{\sigma\tau})(\partial_\rho g_{\tau\nu}) + (\mu \leftrightarrow \nu) \right] \\
&= -\frac{1}{2} g^{\lambda\rho} g_{\mu\nu,\lambda\rho} + \frac{1}{2} g^{\lambda\rho} g^{\sigma\tau} \left[g_{\mu\sigma,\lambda} g_{\nu\tau,\rho} + g_{\nu\sigma,\lambda} g_{\mu\tau,\rho} \right] \\
&= -\frac{1}{2} g^{\lambda\rho} g_{\mu\nu,\lambda\rho} + g^{\lambda\rho} g^{\sigma\tau} g_{\mu\sigma,\lambda} g_{\nu\tau,\rho}.
\end{aligned} \tag{4.19}$$

We thus arrive at the expression

$$R_{\mu\nu} = -\frac{1}{2} g^{\lambda\rho} \partial_\lambda \partial_\rho g_{\mu\nu} + g^{\lambda\rho} g^{\sigma\tau} \partial_\lambda g_{\mu\sigma} \partial_\rho g_{\nu\tau} - \Gamma_{\mu\rho\sigma} \Gamma_{\nu}{}^{\rho\sigma} \tag{4.20}$$

for the Ricci tensor in harmonic gauge.

Notably, second order derivatives only appear in the first term, with the Lorentzian signature of the (inverse) metric making the vacuum Einstein equations $R_{\mu\nu} = 0$ hyperbolic *in this gauge*. More specifically, in this gauge, the Einstein equations take the form of a quasilinear second-order hyperbolic PDE, with non-linear terms quadratic in first derivatives. Hence, for "weak" fields—that is, weak perturbations away from Minkowski space—the second-order term dominates the metric's dynamics, effectively turning it into a wave-like equation. At larger field strengths, the terms quadratic in first-order derivatives start modifying the dynamics away from pure wave propagation.

Schematically, we may write the vacuum Einstein equations in harmonic gauge as

$$g^{\rho\sigma} \partial_\rho \partial_\sigma g_{\mu\nu} = F_{\mu\nu}(g, \partial g), \tag{4.21}$$

for $F_{\mu\nu}(g, \partial g)$ a collection of functions depending at most quadratically on ∂g. This insight will be central for applying local existence and uniqueness results in upcoming sections, in which we will refer to the equation above as the *reduced* Einstein equations (in harmonic gauge).

4.2 Aside: Dynamics of Lapse and Shift

Before moving on to the local and global analysis of solutions to Einstein's equations, let us consider the connection between the harmonic gauge and the metric in terms of ADM variables. Recall that we had identified the lapse α and the shift β to be *pure gauge* variables. Imposing the harmonic gauge condition,

$$\Box_g x^{\bar{\mu}} = 0, \tag{4.22}$$

fixes the gauge, and should hence impose conditions on the lapse and shift. In the following, we analyse the nature of these conditions.

Recalling that

$$\Box_g x^{\bar\mu} = -g^{\rho\sigma}\Gamma^{\bar\mu}{}_{\rho\sigma} =: -\Gamma^{\bar\mu}, \tag{4.23}$$

we note that the contracted Christoffel symbols $\Gamma^{\bar\mu}$ may be calculated as

$$\Gamma^{\bar\mu} = -\Box_g x^{\bar\mu} = -\frac{1}{\sqrt{g}}\partial_\nu(\sqrt{g}g^{\nu\lambda}\underbrace{\partial_\lambda x^{\bar\mu}}_{=\delta^{\bar\mu}{}_\lambda}) = -\frac{1}{\sqrt{g}}\partial_\nu(\sqrt{g}g^{\nu\bar\mu}) \tag{4.24}$$

We remind ourselves that the harmonic gauge condition amounts to requiring $\Gamma^\mu = 0$. Besides giving us a straightforward way of establishing a relationship between the harmonic gauge and the lapse and shift, this informs us of another, more fundamental fact: Even though the original expression $\Box_g x^{\bar\mu} = 0$ might appear to be a condition on the coordinates, it can equivalently be seen as a condition on the metric components. This viewpoint will be useful in the following when we consider how to impose initial conditions for the system (4.21), which, a priori, requires us to specify all of the components $g_{\mu\nu}, \partial_t g_{\mu\nu}$ on some Cauchy surface.

To find equations for the lapse and shift imposed by the harmonic gauge condition, we need to evaluate Γ^t, Γ^i explicitly in terms of the ADM variables. Beginning with the former, we evaluate

$$\begin{aligned}
\Gamma^t &= -\frac{1}{\alpha\sqrt{\gamma}}\partial_\nu(\alpha\sqrt{\gamma}g^{\nu t}) = -\frac{1}{\alpha\sqrt{\gamma}}\partial_t(\alpha\sqrt{\gamma}g^{tt}) - \frac{1}{\alpha\sqrt{\gamma}}\partial_i(\alpha\sqrt{\gamma}g^{it}) \\
&= \frac{1}{\alpha\sqrt{\gamma}}\partial_t\left(\frac{\sqrt{\gamma}}{\alpha}\right) - \frac{1}{\alpha\sqrt{\gamma}}\partial_i\left(\frac{\sqrt{\gamma}}{\alpha}\beta^i\right) \\
&= -\frac{1}{\alpha^3}\partial_t\alpha + \frac{1}{\alpha^2}\underbrace{\frac{1}{\sqrt{\gamma}}\partial_t\sqrt{\gamma}}_{=\frac{1}{2}\gamma^{k\ell}\partial_t\gamma_{k\ell}} + \frac{1}{\alpha^3}\beta^i\partial_i\alpha - \frac{1}{\alpha^2}\underbrace{\frac{1}{\sqrt{\gamma}}\partial_i(\sqrt{\gamma}\beta^i)}_{=\bar\nabla_i\beta^i=\gamma^{k\ell}\bar\nabla_{(k}\beta_{\ell)}} \\
&= -\frac{1}{\alpha^3}\Big(\underbrace{\partial_t\alpha - \beta^i\partial_i\alpha}_{\alpha n[\alpha]} - \frac{\alpha}{2}\gamma^{k\ell}\underbrace{\left(\partial_t\gamma_{k\ell} - \bar\nabla_k\beta_\ell - \bar\nabla_\ell\beta_k\right)}_{=-2\alpha K_{k\ell}}\Big) \\
&= -\frac{1}{\alpha^2}\big(n[\alpha] + \alpha K\big).
\end{aligned} \tag{4.25}$$

Here, $\bar\Gamma^i{}_{jk}$ denotes the three-dimensional Christoffel symbols associated to the spatial metric γ_{ij}, and we have

$$n[f] = \frac{1}{\alpha}(\partial_t - \beta^i\partial_i)f \tag{4.26}$$

A similar(ly straightforward but cumbersome) derivation reveals that (cf. e.g. [4], Appendix B)

$$\Gamma^i = -\frac{1}{\alpha}\big(n[\beta^i] + \gamma^{ij}\partial_j\alpha\big) + \bar\Gamma^i - \beta^i\Gamma^t, \tag{4.27}$$

with $\bar\Gamma^i = \gamma^{k\ell}\bar\Gamma^i{}_{k\ell}$ denoting the foliation-intrinsic contracted Christoffel symbols.

Imposing the harmonic gauge condition $\Gamma^\mu = 0$ allows us to solve for $n[\alpha]$ and $n[\beta^i]$, which yields

$$n[\alpha] = -\alpha K, \tag{4.28}$$

$$n[\beta^i] = \alpha\bar\Gamma^i - \gamma^{ij}\partial_j\alpha. \tag{4.29}$$

This is a set of sourced transport equations for the lapse α and the components of the shift, β^i, along the normal flow of the foliation. This tells us that imposing the harmonic gauge condition promotes the a priori non-dynamical lapse and shift to dynamical variables. In light of the remark that ADM with a fixed choice for the lapse and shift functions—such as $\alpha = 1, \beta = 0$—is ill-posed, this is expected; If the harmonic gauge condition is to yield a well-posed formulation of the Einstein equations, it should not fix α, β to specific functions, but rather, make them dynamical—which is precisely what happens, as we just found.

4.3 Initial Data and Gauge Consistency

To conclude this section, let us briefly address an apparent discrepancy in the specification of initial data for equation (4.21) and the ADM-harmonic-gauge system. For the former, second-order quasilinear wave-like equation, initial conditions are specified by fixing $g_{\mu\nu}$ and $\partial_t g_{\mu\nu}$ on some initial Cauchy surface. Given that both these quantities are symmetric, this amounts to a total of 20 functions to be specified. From the ADM perspective however, with lapse and shift promoted to dynamical variables through the harmonic gauge condition, we must specify (γ, K) (subject to the constraints) and α, β^i—but not $\partial_t\alpha, \partial_t\beta^i$—on the initial Cauchy surface. The initial time derivatives of the lapse and shift do not need to be specified as their transport equations (4.28) and (4.29) are only of first order. This amounts to only 16 functions, which are subject to the four constraints (3.9), yielding a final total of 12 functions to be prescribed as initial data on the Cauchy surface. This mismatch in the count of degrees of freedom might make it appear that these two formulations of general relativity are incompatible—a disastrous conclusion. Fortunately, however, this is not the case. To see this, we should remind ourselves that solutions to equation (4.21) are only valid solutions to the full Einstein equations *if the harmonic gauge condition holds*. Further, the initial data for equation (4.21) is still subject to general relativity's constraints; We are thus not free to pick the initial data for $g_{\mu\nu}$ and $\partial_t g_{\mu\nu}$ completely arbitrarily, but must ensure that they satisfy both the physical and gauge constraints. This reduces the degrees of freedom appropriately.

A constructive way of thinking about this is as follows: One can specify the initial data first in the ADM-harmonic-gauge formulation in the form of $(\alpha, \beta^i, \gamma_{ij}, K_{ij})$, with (γ, K) subject to the Hamiltonian and momentum constraints—in this formulation, it is very clear what the dynamical degrees of freedom are. To then populate the initial data for $g_{\mu\nu}$ and $\partial_t g_{\mu\nu}$, we can use the transport equations (4.28) and (4.29) to obtain $\partial_t\alpha$ and $\partial_t\beta^i$. From that point onwards, it is straightforward to evaluate the initial conditions for

$$[g_{\mu\nu}] = \begin{pmatrix} -\alpha^2 + \gamma_{k\ell}\beta^k\beta^\ell & \gamma_{jk}\beta^k \\ \gamma_{ik}\beta^k & \gamma_{ij} \end{pmatrix} \tag{4.30}$$

and

$$[\partial_t g_{\mu\nu}] = \begin{pmatrix} -2\alpha\partial_t\alpha + (\partial_t\gamma_{k\ell})\beta^k\beta^\ell + 2\gamma_{k\ell}\beta^k\partial_t\beta^\ell & (\partial_t\gamma_{jk})\beta^k + \gamma_{jk}\partial_t\beta^k \\ (\partial_t\gamma_{ik})\beta^k + \gamma_{ik}\partial_t\beta^k & \partial_t\gamma_{ij} \end{pmatrix} \tag{4.31}$$

from the data provided by $(\alpha, \beta^i, \gamma_{ij}, \partial_t\alpha, \partial_t\beta^i, \partial_t\gamma_{ij})$, with $\partial_t\gamma_{ij}$ obtained from K_{ij}.

There is one more thing to consider here. In Section 3.2, we studied the consistency of the Hamiltonian and momentum constraints—that is, we verified that if they hold for the initial data (γ, K), they are preserved by the evolution governed by

$$G_{ij} = 0. \tag{4.32}$$

We would hope for the harmonic gauge condition to have the same property under the evolution dictated by equation (4.21)—we want to provide initial data satisfying the gauge condition, and obtain a solution to the equation which satisfies it at all times. The fact that this is true is shown in chapter 10 of [6]—let us briefly review the argument here. Similarly to the consistency of the physical constraints (3.9), this follows from the Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$. Concretely, replacing $\Box_g x^{\bar{\mu}}$ by $\Gamma^{\bar{\mu}}$ in equation (4.11) and constructing $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$, one derives that

$$\nabla^\mu G_{\mu\nu} = 0 \quad \Longleftrightarrow \quad g^{\rho\sigma}\partial_\rho\partial_\sigma\Gamma^\mu + (\text{terms linear in } \partial_\sigma\Gamma^\mu \text{ and } \Gamma^\mu) = 0. \tag{4.33}$$

Given the initial condition $\Gamma^\mu = \partial_t\Gamma^\mu = 0$ on the initial data surface, we have the unique solution $\Gamma^\mu = 0$. This establishes that further to the physical constraints, the harmonic gauge condition is also propagated under evolution. Thus, importantly, a solution to the reduced vacuum Einstein equations in harmonic gauge, equation (4.21), is a solution to the full vacuum Einstein equations provided that its initial data satisfies both the physical and gauge constraints.

5 Existence of the Unique Maximal Development

Now that we have established a geometric and PDE-theoretic perspective of how to view general relativity as a dynamical PDE, we can move on to the study of existence and uniqueness of solutions. To this end, we will first introduce some notions that will enable us to talk more effectively about what we mean by "solutions" (so-called *developments*) as well as their *extensions*, *maximality* and *uniqueness*. We then consider results which provide us with local existence and uniqueness, which are the primary input into the main result we aim to prove, the Choquet-Bruhat and Geroch theorem [1] (1969). This theorem establishes the existence and uniqueness of a maximal development of initial data. One of the core tools used in its proof is Zorn's Lemma; For the benefit of the reader, and perhaps to address the author's own need to revisit the distant memories of introductory set theory—we also review its underlying definitions and axiomatic context in the following.

5.1 Definitions

The starting point for a solution to any dynamical PDE is its initial data. In the general language of PDEs, this initial data is unconstrained. In the following, however, we will always consider initial data which satisfies the Hamiltonian and momentum constraints (3.9)—for this reason, we extend the definition of initial data to reflect this.

Definition 5.1. *(Initial Data) Let Σ be a three-dimensional manifold and γ a Riemannian metric on Σ. Further, let be K a symmetric rank-2 tensor field on Σ. We call the tuple (Σ, γ, K) an* initial data set *or simply* initial data *if the constraint equations*

$$\bar{R} + K^2 - K_{ij}K^{ij} = 0, \qquad \bar{\nabla}_i(K^{ij} - \gamma^{ij}K) = 0 \tag{5.1}$$

hold, where $K = \gamma^{ij}K_{ij}$ is the trace of K with respect to γ.

From context, it should be clear that the symmetric rank-2 tensor field K in the above definition represents the extrinsic curvature of the initial data. It is important to note, however, that in the context of an initial data set (Σ, γ, K), this interpretation does not make sense as is—a manifold cannot have extrinsic curvature without being embedded into another, ambient manifold. It is for this reason that K is merely required to be a symmetric rank-2 tensor field on Σ—properties that the extrinsic curvature of any embedded manifold has. Because of this, it is up to the definition of a "solution", or more precisely, a *development* of the initial data to ensure that (Σ, γ, K) embeds correctly:

Definition 5.2. *(Development) Let (Σ, γ, K) be initial data. We call a four-dimensional Lorentzian manifold $(\mathcal{M}, g)$ a* development *of (Σ, γ, K) if the following hold:*

(i) $(\mathcal{M}, g)$ solves the vacuum Einstein equations,

$$R_{\mu\nu}[g] = 0 \quad \text{on } \mathcal{M}, \tag{5.2}$$

(ii) there exists a smooth embedding $\iota : \Sigma \to \mathcal{M}$,

(iii) the hypersurface $\iota(\Sigma) \subseteq \mathcal{M}$ is a Cauchy surface of $(\mathcal{M}, g)$,

(iv) and the embedding $\iota : \Sigma \to \mathcal{M}$ is isometric and respects the prescribed extrinsic curvature, that is,

$$\iota^* g = \gamma, \quad -\frac{1}{2}\iota^*(\mathcal{L}_n g) = K, \tag{5.3}$$

where n denotes the timelike future-directed unit normal vector field of $\iota(\Sigma) \subset \mathcal{M}$.

This definition encapsulates the two important aspects of a "solution provided initial data" in a geometric manner: it demands that $(\mathcal{M}, g)$ solves the vacuum Einstein equations, making it a physical spacetime, and requires the specified initial data (Σ, γ, K) to be embedded correctly into the ambient spacetime $(\mathcal{M}, g)$—that is, isometrically and with the extrinsic curvature specified by K. The requirement that $\iota(\Sigma)$ be a Cauchy surface of $(\mathcal{M}, g)$ makes the spacetime globally hyperbolic, an assumption we justified to be necessary for uniqueness results to hold in Section 2. Consequently—and loosely speaking—$\mathcal{M}$ can at most be "as large" as $D[\iota(\Sigma)]$, the domain of dependence.[1]

We should note that this definition is such that the distinction between (Σ, γ, K) and its embedding $\iota(\Sigma) \subseteq \mathcal{M}$ can, for the most part, be neglected. We may hence consider $\Sigma \subseteq \mathcal{M}$ to be a subset in most contexts, but should keep this technicality in mind in situations where (Σ, γ, K) serves as the initial data for two distinct developments $(\mathcal{M}, g)$ and $(\mathcal{M}', g')$.

Remark 5.1. In literature more modern than [1], such as e.g. the constructive "de-zornified" version of the Choquet-Bruhat and Geroch proof given by Sbierski [2], the notion of a development is typically referred to as a *globally hyperbolic development*, abbreviated as *GHD*. Since this essay does not involve any other notions of developments, we opt to follow [1] in their use of the term *development* for purposes of readability.

Though there exists a general notion of extensions for spacetime manifolds, in the following, we will require a more specialised definition for the extension of developments, for which we reserve the term *extension*. For clarity, in the following, we refer to the more general notion of extensions that exists for arbitrary spacetime manifolds as *spacetime extension*.

Definition 5.3. *Let $(\mathcal{M}, g)$ be a development of initial data (Σ, γ, K). We call $(\widetilde{\mathcal{M}}, \tilde{g})$ an* extension *of $(\mathcal{M}, g)$ if it is a development of the same initial data and there exists an open subset $U \subseteq \widetilde{M}$ containing the embedding of Σ such that $(U, \tilde{g}|_U)$ and $(\mathcal{M}, g)$ are isometric. In this case, we refer to $(\mathcal{M}, g)$ as a* subdevelopment *of $(\widetilde{\mathcal{M}}, g)$.*

To clarify further the distinction between the use of the terms *extension* and *spacetime extension* in the following, we note that an extension of a development $(\mathcal{M}, g)$ of some initial data (Σ, γ, K) is itself a development of the same initial data (and by that, globally hyperbolic). A spacetime extension of $(\mathcal{M}, g)$, on the other hand, need not be a development, and is not required to embed Σ as a Cauchy surface.

Again—similar to the interplay of initial data and a development thereof—this definition is such that in most contexts, we can consider a development $(\mathcal{M}, g)$ which is extended by $(\widetilde{\mathcal{M}}, \tilde{g})$ to be a subset of its extension, allowing us to write $\mathcal{M} \subseteq \widetilde{\mathcal{M}}$. This does however demand more careful attention when dealing with two different extensions of the same development.

As our last two definitions for this section, we should introduce the notion of a common subdevelopment and specify what it means for a development to be *maximal*:

Definition 5.4. *(Common Subdevelopment) We say that a development $(\mathcal{M}, g)$ is a* common subdevelopment *of two developments $(\mathcal{M}_1, g_1)$ and $(\mathcal{M}_1, g_2)$ if it is a subdevelopment of both. Here, all three developments are taken to be of the same initial data (Σ, γ, K).*

Definition 5.5. *(Maximal Development) Let $(\mathcal{M}, g)$ be a development of initial data (Σ, γ, K). We call $(\mathcal{M}, g)$ a* maximal development *or simply* maximal *if $(\mathcal{M}, g)$ admits no extension other than itself.*

[1] This statement only makes sense on intuitive grounds, as of course, rigorously speaking, $D[\iota(\Sigma)]$ is always defined as a subset of $(\mathcal{M}, g)$, making it a tautology. The way to interpret the intuition is that when thinking of $(\mathcal{M}, g)$ as being extended by some larger Einstein manifold $(\widetilde{\mathcal{M}}, g)$, then $(\mathcal{M}, g)$ can be at most as large as $D[\iota(\Sigma)]$ as defined in the extension $(\widetilde{\mathcal{M}}, g)$.

In other words, a maximal development is a development which cannot be extended further. The Choquet-Bruhat–Geroch theorem, roughly speaking, states that any initial data (Σ, γ, K) admits a maximal extension which is unique in the sense that it is an extension of any other development. A priori, a maximal extension must not be unique; this is very succinctly explained by Choquet-Bruhat and Geroch in the introduction of [1]:

> "A priori, it might be possible that, once the solution has been integrated beyond a certain point in some region, the option, previously available, of further evolution in some quite different region has been destroyed."

Put differently, it could be that at some point in the evolution of the spacetime, we hit a "fork in the road", and must decide whether to continue evolving the spacetime to maximality into one direction or the other, giving us multiple maximal developments. Proving that this is not the case will be the culmination of this essay.

5.2 Local Results

Although we have set up a comparably large amount of geometric and PDE-theoretic background in the previous sections, the input into the main Choquet-Bruhat–Geroch theorem is remarkably lean. As presented in [1] in geometric language, it boils down to the following two results. The first is a local existence result:

Theorem 5.1. *Provided initial data (Σ, γ, K), there exists a development $(\mathcal{M}, g)$.*

We note that this result does not provide us with any uniqueness or guarantee on "how far" we may evolve the initial data—only that there exists a development, however small. The second result provides us with local uniqueness:

Theorem 5.2. *Any two developments of initial data (Σ, γ, K) are extensions of a common subdevelopment.*

Intuitively speaking, this theorem tells us that if we consider two distinct developments of the same initial data (Σ, γ, K), they must agree on some open neighbourhood of Σ.

Following [8], let us outline how these results come about from the theory we have established so far. As mentioned before, the reduced Einstein equations in harmonic gauge,

$$g^{\rho\sigma}\partial_\rho\partial_\sigma g_{\mu\nu} = F_{\mu\nu}(g, \partial g), \tag{5.4}$$

are a quasilinear second-order hyperbolic PDE system for the components $g_{\mu\nu}$. Standard PDE theory, which we will treat as a blackbox for the purposes of this essay, provides us with the existence of a unique solution (in the PDE sense, by equality of functions) for initial data $(g_{\mu\nu}|_{t=0}, \partial_t g_{\mu\nu}|_{t=0})$ on (a subset of) the domain of dependence of the initial data. Concretely, this means the following:

Proposition 5.1. *Given sufficiently smooth initial data $(g_{\mu\nu}|_{t=0}, \partial_t g_{\mu\nu}|_{t=0})$ on some open neighbourhood $U \subset \mathbb{R}^3$, there exists a unique solution of equation (5.4) on an open neighbourhood of $\{t = 0\} \times U \subset \mathbb{R} \times U$.*

So far, this is merely a statement about solutions of equation (5.4), with no geometric meaning attached to them. Translating this result into the geometric language of Theorems 5.1 and 5.2—which deals with isometries instead of functional equality—requires some additional work.

Proof. (Sketch, of Theorem 5.1) Given (geometric) initial data (Σ, γ, K), we should construct appropriate (PDE) initial data $(g_{\mu\nu}|_{t=0}, \partial_t g_{\mu\nu}|_{t=0})$, where for simplicity we assume that Σ embeds into the spacetime manifold as the $\{t = 0\}$ surface. Following our previous discussion in Section 4.3, we may do so by choosing coordinates on Σ and picking an initial lapse α and shift β, recovering $\partial_t \gamma_{ij}$ from K_{ij}. Moreover, the functions $\partial_t \alpha$ and $\partial_t \beta^i$ are fixed by the first-order gauge transport equations (4.28) and (4.29). The formulae (4.30) and (4.31) then allow us to determine the initial data $(g_{\mu\nu}|_{t=0}, \partial_t g_{\mu\nu}|_{t=0})$, to which a unique local solution $g_{\mu\nu}$ to equation (5.4) is associated by Proposition 5.1.

As we have seen before, both the physical constraints (3.9) and the harmonic gauge $\Gamma^\mu = 0$ are *consistent* under the evolution of the reduced Einstein equations in harmonic gauge, equation (5.4). Thus, the obtained $g_{\mu\nu}$ solves the *full* Einstein vacuum equations, $R_{\mu\nu} = 0$, and we obtain a spacetime $(\mathcal{M}, g)$. Clearly, Σ embeds smoothly into $(\mathcal{M}, g)$ as a Cauchy surface, being the $\{t = 0\}$ surface. Moreover, it embeds with the correct induced metric and extrinsic curvature, given how we constructed $(g_{\mu\nu}|_{t=0}, \partial_t g_{\mu\nu}|_{t=0})$ from (Σ, γ, K). □

Let us now move on to geometric uniqueness, Theorem 5.2.

Proof. (Sketch, of Theorem 5.2) Since we already have local "PDE-type" uniqueness for solutions to equation (5.4), we would like to translate this into geometric language. To this end, suppose we have two developments, $(\mathcal{M}_1, g_1)$ and $(\mathcal{M}_2, g_2)$, of the same initial data (Σ, γ, K). Denoting the isometric embeddings of the initial data by

$$\iota_1 : \Sigma \to \mathcal{M}_1, \quad \iota_2 : \Sigma \to \mathcal{M}_2, \tag{5.5}$$

we obtain an isometry $\psi : \iota_2 \circ (\iota_1)^{-1} : \mathcal{M}_1 \supseteq \iota_1(\Sigma) \to \iota_2(\Sigma) \subseteq \mathcal{M}_2$, relating "equivalent" points on the embedded initial data $\iota_1(\Sigma)$ and $\iota_2(\Sigma)$. Picking coordinates on $\iota_1(\Sigma)$, the maps ψ and $(\iota_1)^{-1}$ provide us with coordinate systems on $\iota_2(\Sigma)$ and Σ. Since ψ is an isometry, the components of $\iota_1^* g_1$ and $\iota_2^* g_2$ match in these coordinates, and we may conclude

$$\gamma_{ij} = (\iota_1^* g_1)_{ij} = (\iota_2^* g_2)_{ij}. \tag{5.6}$$

Further, since ι_1, ι_2 respect the extrinsic curvature, we additionally obtain

$$K_{ij} = -\frac{1}{2} \iota_1^* (\mathcal{L}_{n_1} g_1)_{ij} = -\frac{1}{2} \iota_2^* (\mathcal{L}_{n_2} g_2)_{ij}. \tag{5.7}$$

We may extend the coordinates on $\iota_1(\Sigma) \subseteq \mathcal{M}_1$ and $\iota_2(\Sigma) \subseteq \mathcal{M}_2$ to harmonic coordinates in respective neighbourhoods, with the freedom of fixing the lapse α and shift β^i on $\iota_1(\Sigma)$ and $\iota_2(\Sigma)$, e.g. to the standard values

$$\alpha = 1, \quad \beta = 0 \quad \text{on } \Sigma. \tag{5.8}$$

By the relationship to the PDE-initial data $(g_{\mu\nu}|_{t=0}, \partial_t g_{\mu\nu}|_{t=0})$, we conclude that in these coordinates, $(g_1, \partial_t g_1)|_{t=0}$ and $(g_2, \partial_t g_2)|_{t=0}$ are equal as functions. Since $(\mathcal{M}_1, g_1)$ and $(\mathcal{M}_2, g_2)$ are both developments, which we just cast into harmonic coordinates, they both solve equation (5.4), with the same initial data. By the PDE-type local uniqueness, we conclude that in these coordinates, $(g_1)_{\mu\nu} = (g_2)_{\mu\nu}$ on some open coordinate neighbourhood of the Cauchy surface, showing that $(\mathcal{M}_1, g_1)$ and $(\mathcal{M}_2, g_2)$ are extensions of a common subdevelopment. This establishes Theorem 5.2. □

5.3 Choquet-Bruhat and Geroch Theorem

The main theorem proven by Choquet-Bruhat and Geroch in their seminal paper [1] states the following:

Theorem 5.3. *Given initial data (Σ, γ, K), there exists a development $(\mathcal{M}, g)$ of this initial data which is an extension of any other development of the same initial data. This development is unique up to isometry.*

Though not strictly necessary from a logical standpoint, we break this statement down into the three results below. This makes its proof more accessible by contributing to the transparency of its logic, and makes clearer how and where precisely Zorn's lemma is invoked.

Proposition 5.2. *Provided initial data (Σ, γ, K), there exists a maximal development.*

This is where Zorn's lemma is featured most prominently. To show the maximal development's uniqueness, we further need the following proposition as a tool to combine two developments:

Proposition 5.3. *Any two developments $(\mathcal{M}, g)$, $(\mathcal{M}', g')$ have a unique common subdevelopment.*

This proposition turns out to be central to proving the final uniqueness result below.

Proposition 5.4. *The maximal development from Proposition 5.2 is an extension of any other development of the same initial data.*

Though at first, this last proposition might not look like a uniqueness result, the fact that the maximal development is unique follows from it rather quickly. Supposing that there are two maximal developments, $M = (\mathcal{M}, g)$, $M' = (\mathcal{M}', g')$, it follows from Proposition 5.4 that both M and M' are extensions of any development, and in particular, of each other. If M extends M' and vice versa, it is clear that they must be equal (up to isometry).

5.3.1 Review: Set Theory and Zorn's Lemma

This section follows [9] to give a brief overview of Zorn's lemma and the set-theoretic concepts surrounding it, putting them into the context in which we will need them in the following.

Ultimately, Zorn's lemma is a statement about the existence of some "largest" or "maximal" element of a given set. To be able to make any such statement, we need a notion of comparison or *order* of the elements within the set—otherwise, the term "largest" becomes hard to make sense of. This comes in the form of a *partial order*, which we introduce now.

Definition 5.6. *(Partial Order) A* partial order *on a set X is a relation $\leq$ which is*

(i) reflexive*, that is, $x \leq x$;*

(ii) anti-symmetric*, if both $x \leq y$ and $y \leq x$ it follows that $x = y$;*

(iii) transitive*, $x \leq y$ and $y \leq z$ imply $x \leq z$.*

In this case, we call $(X, \leq)$ a partially ordered set.

We note that this definition allows for elements to be incomparable, that is, there might exist pairs $x, y \in X$ such that neither $x \leq y$ nor $y \leq x$. Hence, partial order is a weaker notion than the perhaps more familiar *total order*:

Definition 5.7. *(Total Ordering) A partial order $\leq$ on a set X is called a* total order *if any two elements are comparable, that is, for any $x, y \in X$ we have $x \leq y$ or $y \leq x$. Then, $(X, \leq)$ is known as a* totally ordered set.

In the following, we compare developments of the same initial data by whether they extend one another, which clearly cannot be a total order. After all, we could start with a development and extend it in two different places, yielding two developments which do not extend one another and are hence incomparable. One can, however, convince oneself that this relation, more formally defined as

$$(\mathcal{M}, g) \leq (\mathcal{M}', g') \quad :\Longleftrightarrow \quad (\mathcal{M}', g') \text{ is an extension of } (\mathcal{M}, g), \tag{5.9}$$

does define a *partial* order on the set of developments of fixed initial data (Σ, γ, K), which we denote by $\mathcal{X}$ in the following.

Now that we have a notion of order or "size", we can define what it means for an element to be maximal:

Definition 5.8. *(Maximal Element) Let $(X, \leq)$ be a partially ordered set. An element $M \in X$ is called* maximal *if $M' \geq M$ implies $M' = M$ for any $M' \in X$.*

In other words, a maximal element of a partially ordered set is an element for which there exists no strictly larger element. However, since elements are allowed to be incomparable, it is possible that X has multiple maximal elements.

For the specific case of $(\mathcal{X}, \leq)$, the partially ordered set of developments of fixed initial data, this turns out not to be the case—in the upcoming sections, we show Choquet-Bruhat's and Geroch's result that a maximal element exists, and that it is unique in the sense that it extends any other development in $\mathcal{X}$ [1]. Here, we note that maximality with respect to $\leq$ is equivalent to the notion of maximality of developments we defined in Definition 5.5.

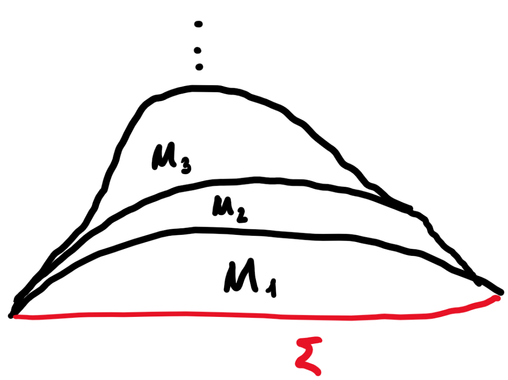
The proof of the existence of such a maximal development makes direct use of *Zorn's lemma*, which we introduce now. To this end, we require two more notions:

Definition 5.9. *(Chain) We call a subset $S \subseteq X$ of a partially ordered set $(X, \leq)$ a* chain *if $(S, \leq)$ with the relation $\leq$ restricted to S is totally ordered.*

Although a chain can be very large—for example, uncountable—it is often instructive to think of a chain as a set of elements $M_i \in X$ for which

$$M_1 \leq M_2 \leq M_3 \leq \ldots \tag{5.10}$$

In $(\mathcal{X}, \leq)$, a chain corresponds to a collection of developments which extend one another progressively, which may be represented diagramatically as



(5.11)

Nevertheless, we should keep in mind that chains are allowed to be uncountable, which introduces some technicalities in the proofs that follow.

Definition 5.10. *(Upper Bound) Given a subset S of a partially ordered set $(X, \leq)$, we call $M \in X$ an* upper bound *of S if $M' \leq M$ for all $M' \in S$.*

We note here that an upper bound of a subset does not necessarily have to be contained in the subset, e.g. the interval $S = (0,1)$ as a subset of $X = \mathbb{R}$ with the standard total order has upper bound 1, even though $1 \notin S$.

We are now ready to state Zorn's lemma:

Lemma 5.1. *(Zorn's Lemma) If every chain in a non-empty partially ordered set $(X, \leq)$ has an upper bound, then X contains a maximal element.*

The usefulness of this result might at first seem somewhat opaque, but is actually rather powerful. This is because we obtain the existence of a maximal element of some partially ordered set simply by considering its totally ordered subsets, which are simpler to deal with. In the concrete case of $(\mathcal{X}, \leq)$, the total order of a chain allows us to "glue together" its elements into a development which extends all the elements of the chain and hence serves as an upper bound. In fact, this is the strategy we employ in the following to prove the existence of a maximal development.

Though powerful, Zorn's lemma has one important drawback. It is equivalent to the controversial *axiom of choice*, whose place in fundamental axiomatic systems is debated. We will return to comment on this further in a later section.

5.3.2 Existence of a Maximal Development

In this section we prove Proposition 5.2—that is, we demonstrate the existence of a maximal development given initial data (Σ, γ, K). Before carrying out this proof, let us give a brief outline of it. As we have remarked before, the existence of a maximal development of (Σ, γ, K) is equivalent to the existence of a maximal element in the set $\mathcal{X}$ of all developments of (Σ, γ, K) with respect to the partial order $\leq$ defined by extension. We may thus prove its existence using Zorn's lemma, which requires us to show that any chain has an upper bound. Concretely, for $(\mathcal{X}, \leq)$, a chain is a set of developments which extend one another progressively. An upper bound of such a chain is an extension of every element of the chain, so we may naturally construct one by "gluing" all developments in the chain into one. We achieve this by taking their disjoint union and identifying points related by an isometry. To verify that the resulting topological space is an upper bound, we have to show that it can be endowed with the structure of a Lorentzian manifold satisfying all properties in Definition 5.2 of a development, and that it extends any element of the chain.

Proof. (Of Proposition 5.2) By Theorem 5.1, there exists a development of (Σ, γ, K), implying that $\mathcal{X}$ is non-empty. Let $\{M_\alpha\} \subseteq \mathcal{X}$ be a chain, with $M_\alpha = (\mathcal{M}_\alpha, g_{(\alpha)})$ developments of (Σ, γ, K) indexed by $\alpha \in \mathcal{A}$ for some (potentially uncountable) index set $\mathcal{A}$. We define a topological space $\widetilde{\mathcal{M}}$ by the disjoint union

$$\widetilde{\mathcal{M}} = \bigsqcup_{\alpha \in \mathcal{A}} \mathcal{M}_\alpha \tag{5.12}$$

Since $\{M_\alpha\}$ is a chain, for any $\alpha, \beta \in \mathcal{A}$, without loss of generality we have $M_\alpha \leq M_\beta$, implying that there exists an isometry $\psi_{\beta\alpha} : \mathcal{M}_\alpha \to \psi_{\beta\alpha}(\mathcal{M}_\alpha) \subseteq \mathcal{M}_\beta$ which leaves the embedded initial data invariant. In particular, $\psi_{\gamma\beta} \circ \psi_{\beta\alpha} = \psi_{\gamma\alpha}$, $\alpha, \beta, \gamma \in \mathcal{A}$. We use these isometries to define an equivalence relation on $\widetilde{\mathcal{M}}$ by setting $p \sim q$ iff

$$\begin{cases} \psi_{\beta\alpha}(p) = q, & \text{if } p \in M_\alpha \leq M_\beta \ni q, \\ \psi_{\alpha\beta}(q) = p, & \text{if } p \in M_\alpha \geq M_\beta \ni q. \end{cases} \tag{5.13}$$

This allows us to define the quotient space $\mathcal{M} := \widetilde{\mathcal{M}}/\sim$. Intuitively speaking, one can imagine the procedure we just carried out as follows: We have multiple sheets of paper (the developments) onto which the same image (the metric structure) is printed, to different levels of completion (how far the initial data was evolved). We can stack these papers onto each other (the disjoint union) in such a way that the same points of the image between the individual papers (those related by isometry) align. Then, we identify these points, gluing and compressing the stack of papers $\widetilde{\mathcal{M}}$ into a single paper $\mathcal{M}$ containing all the image fragments of the individual paper sheets.

We now work our way towards showing that $\mathcal{M}$, equipped with an appropriate choice of metric, is a development, which, by construction, extends any M_α, proving that it is an upper bound. To this end, we observe that by definition of the quotient topology, any $\mathcal{M}_\alpha$ is homeomorphic to some subset of $\mathcal{M}$, so that we may think of them as subsets, $\mathcal{M}_\alpha \subseteq \mathcal{M}$, rather than being embedded into some complex quotient topology. Firstly, we should verify that $\mathcal{M}$ can be endowed with the structure of a manifold:

- $\mathcal{M}$ is clearly Hausdorff: given any two points $p, q \in \mathcal{M}$, there exists an $\alpha \in \mathcal{A}$ such that $p, q \in \mathcal{M}_\alpha$. Since $\mathcal{M}_\alpha$ is Hausdorff, there exist disjoint open subsets $U_p, U_q \subseteq \mathcal{M}_\alpha \subseteq \mathcal{M}$ with $p \in U_p, q \in U_q$, establishing that $\mathcal{M}$ is Hausdorff.

- $\mathcal{M}$ is also second-countable. Showing this is highly technical, and does not add much to geometrical or physical intuition, so we omit this part of the proof. The non-triviality of second-countability of $\mathcal{M}$ comes from the fact that the index set $\mathcal{A}$ may be very large, making it difficult to construct a countable basis of $\mathcal{M}$ given the countable bases of the topologies on $\mathcal{M}_\alpha$. For a rigorous proof as to why $\mathcal{M}$ is second-countable, it is advised to refer to e.g. chapter 23 of [10].

- Similarly to the Hausdorff property, the smooth structure of $\mathcal{M}$ is inherited from that on the $\mathcal{M}_\alpha$. For example, around a point $p \in \mathcal{M}$, we can construct a chart by taking a chart around $p \in \mathcal{M}_\alpha$ (for suitable $\alpha \in \mathcal{A}$) which by the open inclusion $\mathcal{M}_\alpha \subseteq \mathcal{M}$ is a chart in $\mathcal{M}$.

We equip $\mathcal{M}$ with a Lorentzian metric by, at each point, assigning the metric $g_{(\alpha)}$ of some $\mathcal{M}_\alpha$ at that point. This is well-defined by the construction of $\mathcal{M}$—we identified points related by isometry. This assignment of metric makes the embeddings $\mathcal{M}_\alpha \subseteq \mathcal{M}$ isometric. Further, it is clear that $(\mathcal{M}, g)$ satisfies the vacuum Einstein equations, since for any $p \in \mathcal{M}$, there exists an index $\alpha \in \mathcal{A}$ and an open neighbourhood $p \in U \subseteq \mathcal{M}_\alpha \subseteq \mathcal{M}$ such that

$$R_{\mu\nu}[g]|_U = R_{\mu\nu}[g_{(\alpha)}]|_U = 0 \tag{5.14}$$

due to the fact that $(\mathcal{M}_\alpha, g_{(\alpha)})$ solves the vacuum Einstein equations.

It remains to check whether the initial data (Σ, γ, K) embeds into $(\mathcal{M}, g)$ appropriately. Since for any $\alpha \in \mathcal{A}$, M_α is a development, Σ embeds into $\mathcal{M}_\alpha \subset \mathcal{M}$ isometrically and with the appropriate extrinsic curvature, and hence, also into $\mathcal{M}$. We are left to show that $\Sigma \subseteq \mathcal{M}$ is a Cauchy surface. To this end, let $\lambda : I \to \mathcal{M}$ be an inextendible causal curve defined on an open interval I. Fix any $(\mathcal{M}', g') = (\mathcal{M}_\alpha, g_{(\alpha)})$, and consider the restriction

$$\lambda' = \lambda|_{\mathcal{M}'} : I' \to \mathcal{M}', \tag{5.15}$$

with $I' \subseteq \lambda^{-1}(\mathcal{M}')$ a connected component. Since $\mathcal{M}' \subseteq \mathcal{M}$ is open and λ continuous, it follows that I' is an open interval, which we denote by $I' = (a, b)$. For a contradiction, we now assume that λ' were not inexendible, that is, without loss of generality, it has a future endpoint $p \in \mathcal{M}'$. By definition, this means that

$$\forall \text{ neighbourhoods } U \subseteq \mathcal{M}' \text{ of } p, \quad \exists t_0 \in I' \text{ such that } \lambda'(t) \in U \quad \forall t > t_0, \tag{5.16}$$

where t is an affine parameter along λ'. Since $\mathcal{M}' \subseteq \mathcal{M}$ is open, p is an interior point of $\mathcal{M}'$. Moreover, by continuity of λ' and the fact that I' is an interval, the future endpoint p must satisfy

$$p = \lim_{t \to b} \lambda'(t). \tag{5.17}$$

However, for this limit of λ' to lie in the interior of $\mathcal{M}$, and $\lambda(t) \notin \mathcal{M}'$ for $t > b$, we must have either one of the following: (i) λ also has a future endpoint at p, (ii) λ has a discontinuity at p, or (iii) I is disconnected. Either case contradicts the assumptions on λ, establishing that λ' is inextendible in $(\mathcal{M}', g')$. Since Σ is a Cauchy surface of $\mathcal{M}' \subseteq \mathcal{M}$, it follows that λ' intersects Σ exactly once, and thus λ intersects Σ at least once. To see that the intersection is unique, suppose that there are two intersections $p_i = \lambda(t_i) \in \Sigma$, $i = 1, 2$, for $t_1 < t_2$. Then, $\lambda|_{[t_1,t_2]}$ is a compact segment of the curve, whence there exists an $\alpha \in \mathcal{A}$ such that $\lambda|_{[t_1,t_2]} \subseteq \mathcal{M}_\alpha$. This is a (potentially extendible) causal curve in $\mathcal{M}_\alpha$ which can be extended into an inextendible causal curve in $\mathcal{M}_\alpha$ that intersects Σ twice—contradicting the fact that Σ is a Cauchy surface of $\mathcal{M}_\alpha$. Thus, λ intersects Σ exactly once, and we have proven that Σ is a Cauchy surface of $\mathcal{M}$.

We have now established that $M = (\mathcal{M}, g)$ is a development of (Σ, γ, K), and thus, $M \in \mathcal{X}$. Moreover, by construction, it is an extension of any M_α in the chain, and by that, an upper bound. By Zorn's Lemma 5.1, we obtain the existence of at least one maximal element of $(\mathcal{X}, \leq)$, corresponding to a maximal development of (Σ, γ, K).

□

5.3.3 Uniqueness of the Maximal Development

This section is dedicated to proving that the maximal development we now know to exist is unique. To this end, we first prove Proposition 5.3, as it is a central result used in proving uniqueness in Proposition 5.4. The proof of Proposition 5.3 is very similar in flavour to that of the existence result from the previous section, Proposition 5.2, allowing us to argue that some parts of the proof follow by analogy. The main difference is that given two developments $M = (\mathcal{M}, g)$ and $M' = (\mathcal{M}', g')$, we will regard a common subdevelopment of M, M' to be a pair (U, ψ) where $U \subseteq \mathcal{M}$ is a subdevelopment of M and $\psi : U \to \psi(U) \subseteq \mathcal{M}'$ is an isometry leaving the initial data (Σ, γ, K) invariant. From this perspective, the construction of an upper bound of a chain becomes simpler, since the chain consists of developments U which are all subsets of $\mathcal{M}$; instead of having to perform the disjoint union and subsequent quotienting by isometry equivalence of points, we can simply take the set union within $\mathcal{M}$. This has as consequence that properties like Hausdorffness and second-countability follow trivially from the induced subspace topology—the isometries ψ, however, need more attention.

Proof. Let $M = (\mathcal{M}, g), M' = (\mathcal{M}', g') \in \mathcal{X}$ be developments. Our goal is to show that there exists a unique maximal common subdevelopment, that is, we can find a maximal subset $U \subseteq \mathcal{M}$ such that $(U, g|_U)$ is a development and there exists an isometry $\psi : U \to \psi(U) \subseteq \mathcal{M}'$.

By Theorem 5.2, M and M' are extensions of some common subdevelopment (U, ψ). Denote by $\mathcal{C}(M, M')$ all such pairs—that is, $\mathcal{C}(M, M')$ is the non-empty collection of all common subdevelopments of M and M'. Our aim is to construct a partial order on $\mathcal{C}(M, M')$ and show that the conditions necessary to invoke Zorn's lemma are satisfied.

To do so, we first show that given two subdevelopments $(U, \psi), (\tilde{U}, \tilde{\psi})$, ψ and $\tilde{\psi}$ agree on the overlap $U \cap \tilde{U}$. To this end, we let $p \in U \cap \tilde{U}$ and show $\psi(p) = \tilde{\psi}(p)$. Let γ be an inextendible timelike geodesic through p, parameterised by proper time. Since M is globally hyperbolic, there exists exactly one intersection of γ and Σ, say $q \in \Sigma \subseteq M$. Assuming without loss of generality that $\gamma(0) = q$, $\gamma(\tau) = p$, the point p is uniquely specified by the pair (γ, τ). Further, since geodesics are fixed uniquely by their initial conditions, γ can be specified uniquely by (q, X), where $X = \dot{\gamma}(0)$ is the initial tangent. Lastly, since γ is assumed to be parameterised by proper time, we have

$$g(\dot{\gamma}, \dot{\gamma}) = -1, \tag{5.18}$$

and therefore, it is sufficient to specify only the projection ΠX of X onto the tangent space of Σ—the normal component can be reconstructed from the normalisation requirement (5.18). Thus, the point p is specified uniquely by the tuple $(q, \Pi X, \tau)$. Correspondingly, the images $\psi(p)$ and $\tilde{\psi}(p)$ are fixed uniquely by the tuples $(\psi(q), \psi_* \Pi X, \tau)$ and $(\tilde{\psi}(q), \tilde{\psi}_* \Pi X, \tau)$, respectively. However, since both ψ and $\tilde{\psi}$ leave Σ fixed, it follows that

$$\psi(q) = \tilde{\psi}(q), \quad \psi_* \Pi X = \tilde{\psi}_* \Pi X \quad \Longrightarrow \quad \psi(p) = \tilde{\psi}(p). \tag{5.19}$$

This has as consequence that

$$(U, \psi) \leq (\tilde{U}, \tilde{\psi}) \quad :\Longleftrightarrow \quad U \subseteq \tilde{U} \tag{5.20}$$

is a well-defined partial order on $\mathcal{C}(M, M')$. In particular, given a chain $\{(U_\alpha, \psi_\alpha) | \alpha \in \mathcal{A}\} \subseteq \mathcal{C}(M, M')$, we may construct an upper bound (U, ψ) by setting

$$U = \bigcup_{\alpha \in \mathcal{A}} U_\alpha, \qquad \psi(p) = \psi_\alpha(p) \text{ for an } \alpha \text{ such that } p \in U_\alpha. \tag{5.21}$$

The fact that $(U, g|_U)$ is a development follows in analogy to the proof of Proposition 5.2; further, by construction, $\psi : U \to \psi(U) \subseteq \mathcal{M}'$ is a well-defined isometry because the isometries ψ_α agree on overlap.

By Zorn's Lemma 5.1, there hence exists a maximal common subdevelopment $(\mathcal{U}, \Psi) \in \mathcal{C}(M, M')$ which cannot be extended further while remaining extendible by both M and M'. This maximal element is unique, since given another maximal element $(\tilde{\mathcal{U}}, \tilde{\Psi})$, their union $\mathcal{U} \cup \tilde{\mathcal{U}}$ would be an extension of $\mathcal{U}$ or $\tilde{\mathcal{U}}$, contradicting their maximality. $\square$

We now come to the proof of the uniqueness result for the maximal development, Proposition 5.4. We first delineate the logic behind it, and then present it in more detail. To show that a maximal development $M = (\mathcal{M}, g)$ of some initial data (Σ, γ, K)—whose existence is provided by Proposition 5.2—extends any other development $M' = (\mathcal{M}', g')$, we make use of the fact that according to Proposition 5.3, they share a unique maximal common development (U, ψ) with $U \subseteq \mathcal{M}$. Again imagining $\mathcal{M}$ and $\mathcal{M}'$ to be sheets of paper, we can "glue" them into a single space, say $\widetilde{\mathcal{M}}$, along the isometrically related regions $U \subseteq \mathcal{M}$ and $\psi(U) \subseteq \mathcal{M}'$. If this space, equipped with the metric inherited by the isometries to form $\widetilde{M} = (\widetilde{\mathcal{M}}, \tilde{g})$, is a development, it is—by construction—sure to extend both M and M'. Since M is maximal, however, it follows that the glued space $\widetilde{M}$ must in fact be equal to M, showing that M extends M', concluding the proof.

In showing that $\widetilde{M}$ is a development, most properties of Definition 5.2 go through without issue. There is, however, one aspect which may fail: the Hausdorff property. To continue the analogy, having glued two sheets of paper together by only applying glue to a subset U (and $\psi(U)$, respectively), the two sheets delaminate outside of U. This can—in principle—cause "non-separable point pairs" in the Hausdorff sense along the boundary of U and $\psi(U)$. A large portion of the proof is devoted to showing that this is not the case, by showing that if it were the case, the maximal common subdevelopment U could be extended to a larger common subdevelopment, contradicting its maximality.

Proof. (Of Proposition 5.4) Let $M = (\mathcal{M}, g)$ denote a maximal development of initial data (Σ, γ, K) and let $M' = (\mathcal{M}', g')$ be any other development. By Proposition 5.3, there exists a unique maximal common subdevelopment $U \subseteq \mathcal{M}'$ and an isometry $\psi : U \to \psi(U) \subseteq \mathcal{M}$ leaving the embedded Σ invariant. We glue M and M' along U, that is, we take a quotiented disjoint union

$$\widetilde{\mathcal{M}} = (\mathcal{M} \sqcup \mathcal{M}')/\sim \tag{5.22}$$

where the quotient is taken with respect to $p \sim q$ iff $\psi(p) = q$ for $p \in U$, $q \in \psi(U)$. The resulting quotient topology on $\widetilde{\mathcal{M}}$, whose quotient map we denote by π, is clearly second countable, since the topologies on $\mathcal{M}$ and $\mathcal{M}'$ are, allowing us to take the (quotient of the) finite union of their countable bases to obtain a countable basis of $\widetilde{\mathcal{M}}$. Further, charts on $\widetilde{\mathcal{M}}$ may be constructed from charts on $\mathcal{M}$ and $\mathcal{M}'$. If $\widetilde{\mathcal{M}}$ were Hausdorff and by that, a manifold, the embeddings $\mathcal{M}, \mathcal{M}' \subseteq \widetilde{\mathcal{M}}$ and the metric structure present on $\mathcal{M}, \mathcal{M}'$ can be used to define $\tilde{g}$ on $\widetilde{\mathcal{M}}$ such that $\widetilde{M} = (\widetilde{\mathcal{M}}, \tilde{g})$ embeds M and M' isometrically. The other properties of a development follow by arguments similar to those presented in the proof of Proposition 5.2. Then, as argued in the outline of this proof's structure, it follows that $\widetilde{M}$ is a development extending M and M', which by maximality of M implies that M extends M'.

We are thus left to prove that $\widetilde{\mathcal{M}}$ is Hausdorff. For a contradiction, suppose it were not. The points of $\widetilde{\mathcal{M}}$ that are "non-Hausdorff" must lie on $\pi(\partial U)$ or $\pi(\partial\psi(U))$, as otherwise it is always possible to find disjoint neighbourhoods of points given the Hausdorff property holding for $\mathcal{M}$ and $\mathcal{M}'$. Since we assumed $\widetilde{\mathcal{M}}$ to be non-Hausdorff, there exist points $p' \in \partial U \subseteq \mathcal{M}'$ and $p \in \partial\psi(U) \subset \mathcal{M}$ such that for any open neighbourhood $\tilde{V} \subseteq \widetilde{\mathcal{M}}$ of $\pi(p')$, we have $\pi(p) \in \tilde{V}$. The lifts of $\tilde{V}$ into $\mathcal{M}$ and $\mathcal{M}'$, the sets

$$V = \pi^{-1}(\tilde{V}) \cap \mathcal{M}, \quad V' = \pi^{-1}(\tilde{V}) \cap \mathcal{M}', \tag{5.23}$$

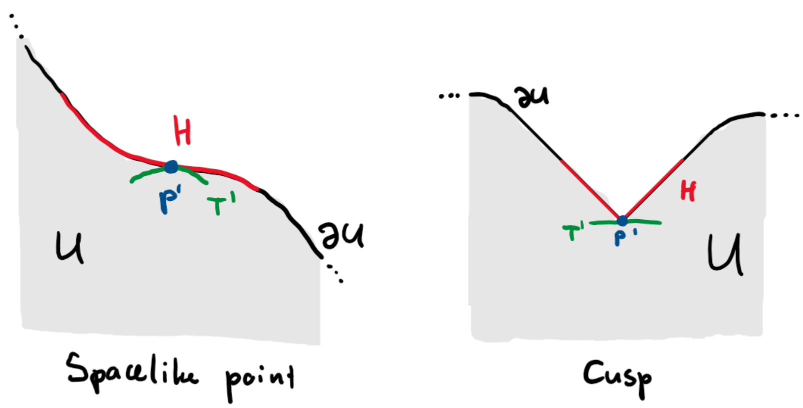
have the property that $\psi(V' \cap U) = V \cap \psi(U)$, whence the failure of the Hausdorff property of $\mathcal{M}$ can be stated as follows:

$$\begin{aligned}&\text{There exist } p \in \mathcal{M} \text{ and } p' \in \mathcal{M}' \text{ such that for any neighbourhood } V' \text{ of } p' \text{ in } \mathcal{M}',\\ &\text{we have } p \in \overline{\psi(V' \cap U)}.\end{aligned} \tag{5.24}$$

We now show that given $p' \in \partial U$, there is at most one $p \in \partial\psi(U)$ with this property. To this end, take such a pair $p' \in \partial U$ and $p \in \partial\psi(U)$, and let γ be a connected timelike curve in U with future endpoint p'. Then, by the isometry property of ψ, the curve $\psi \circ \gamma$ is a connected timelike curve in $\psi(U)$, and by the above characterisation of the pair (p, p'), p is an accumulation point of $\psi \circ \gamma$. Since $p \in \partial\psi(U)$ and $\psi \circ \gamma \subseteq \psi(U) \subseteq \mathcal{M}$, p must be the unique endpoint of $\psi \circ \gamma$. Hence, given $p' \in \partial U$, the associated $p \in \partial\psi(U)$ is unique.

Now, let $H \subseteq \partial U$ be the set of all $p' \in \mathcal{M}'$ for which there exists such a partner $p \in M$. By construction, the Hausdorff property of $\widetilde{\mathcal{M}}$ is equivalent to $H = \emptyset$. One can show that the set H is open in ∂U, and that null geodesics in H have endpoints in H. Intuitively speaking, openness comes about in the "two sheets of paper" analogy that when we have a non-Hausdorff pair at the edge of the glued portion, nearby boundary points are also too close to each other to be separated by open neighbourhoods. The latter property of H is due to the fact that ∂U and $\partial\psi(U)$ are boundaries of globally hyperbolic sets.

These two properties combine into a statement about the causal structure of H: since H is open, any null geodesic in H must leave H into $\mathcal{M}' \setminus \partial U$, as the endpoints cannot lie on the boundary of H in ∂U because H is open in ∂U. This means that H cannot be a null hypersurface, as for a null hypersurface, its generators—the null geodesic integral lines of its normal vector field—run within it. This implies that at least two points, H does not have a null normal vector field; roughly speaking, one is caused by the past and one by the future endpoints in H which null geodsics in H must have—this will become clearer with a sketch below. The failure of a point $p' \in H$ to have a null normal can arise either from the surface being spacelike at that point ($\iota^* g|_{p'}$ Riemannian) or by H failing to be smooth at p' and developing a cusp. The normal vector field, if it exists, cannot be timelike by global hyperbolicity of U, so these are the only two options. We can illustrate these options as follows, for simplicity considering ∂U to be in the future of Σ:



(5.25)

At such points, it is possible to construct a (possibly small) spacelike surface T' such that $T' \setminus \{p'\} \subset U$ and $p' \in H$. We note that the time-reverse of the latter, a "future-pointing cusp", would not allow for such a construction. However, it cannot be that the only point where H admits no null normal is such a future-pointing cusp; this would force null geodesics in H to only have a future but no past endpoint in H. Hence, there must either be a spacelike point or a "past-pointing cusp". In higher dimensions than the $1+1$-dimensional sketches shown above, cusps could be a mixture of future- and past-pointing—requiring interpolation between future- and past-directed null geodesics[2], which necessarily leads to points where H is spacelike.

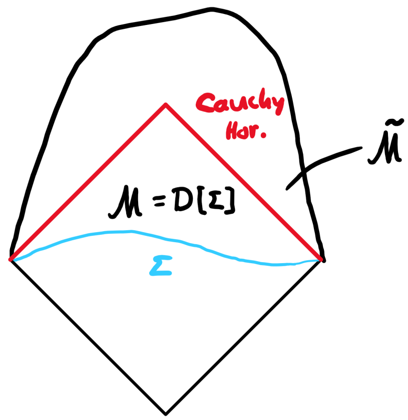
The existence of such a spacelike surface T' with $T' \setminus \{p'\} \subseteq U$ and $p' \in H \subseteq \partial U$ now allows us to obtain a contradiction: Defining $T = \psi(T') \cup \{p\}$, where $p \in \mathcal{M}$ is the unique non-Hausdorff partner of p', we obtain two isometric surfaces $T' \subseteq \mathcal{M}'$ and $T \subseteq \mathcal{M}$. Appropriate open neighbourhoods of T and T' in $\mathcal{M}$ and $\mathcal{M}'$ are developments of the isometric initial data on T and T'—consequently, by Theorem 5.2, they have a common subdevelopment, say $D \subseteq \mathcal{M}'$. Thus, the common development U of $\mathcal{M}$ and $\mathcal{M}'$ may be enlarged to $D \cup U \supsetneq U$, contradicting the fact that U is maximal. Hence, H must be empty and thus $\widetilde{\mathcal{M}}$ is Hausdorff, making it a development, completing the proof. □

6 Conclusion and Outlook

To conclude this essay, let us go over some implications and other aspects of the Choquet-Bruhat–Geroch theorem. To begin, let us interpret the result in more detail. As we have mentioned before, given initial data (Σ, γ, K), we can predict at most a neighbourhood of Σ for which Σ is a Cauchy surface—that is, we can only predict a globally hyperbolic neighbourhood of the initial data. Choquet-Bruhat's and Geroch's result is consistent with this; it informs us that there exists a unique, globally hyperbolic spacetime that embeds (Σ, γ, K) as initial data—the maximal development—but does not make any statement about the contents of any spacetime extension of this development which no longer

[2]when viewed as moving away from p'

constitutes a development:



(6.1)

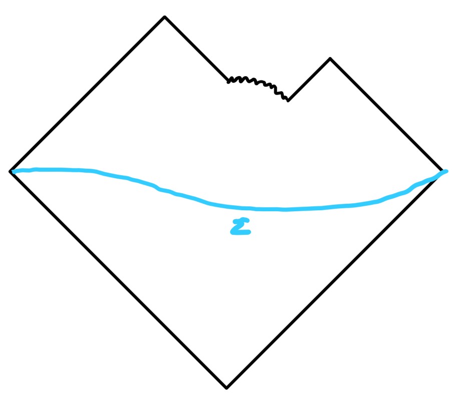
That is to say, if we were to take the unique maximal development $(\mathcal{M}, g)$ of the initial data, and extend it to a larger spacetime $(\widetilde{\mathcal{M}}, \tilde{g})$ in such a way that

$$\mathcal{M} = D[\Sigma] \subsetneq \widetilde{\mathcal{M}}, \tag{6.2}$$

with $D[\Sigma]$ the domain of dependence of Σ as determined in $(\widetilde{\mathcal{M}}, \tilde{g})$, there is no guarantee of uniqueness in the region $\widetilde{\mathcal{M}} \setminus \mathcal{M}$.

However, equation (6.2) tells us that there must be a Cauchy horizon at the boundary $\partial\mathcal{M}$ of $\mathcal{M} \subsetneq \widetilde{\mathcal{M}}$. Recall here that roughly speaking, the strong cosmic censorship conjecture (SCC) states that, given "generic" initial data (Σ, γ, K), its maximal development does not contain any Cauchy horizons. That is, together with the SCC and provided generic initial data, Choquet-Bruhat's and Geroch's result not only gives us the existence of a unique maximally extended globally hyperbolic development of the initial data, but in addition ensures that this is the entire spacetime; In other words, the maximal development is spacetime-inextendible.

Another important aspect to highlight is what the Choquet-Bruhat–Geroch theorem says (or rather, does not say) about the presence of singularities in a spacetime. Even though, given initial data, we are provided with the existence of a unique maximal development, this does not magically get rid of all singularities. After all, the maximality of the development refers to the fact that it cannot be extended further—not that the development has a spacetime diagram corresponding to a perfect diamond as shown above. Beyond just "moving too far away" from the initial data on Σ, another reason why at some point we might not be able to continue extending a development is because we have ran into a singularity. Schematically, this could lead to a spacetime diagram such as



(6.3)

Even though this diagram is not a perfectly complete diamond, it is still *maximal*; we cannot extend it further as a development. Ergo, the Choquet-Bruhat–Geroch theorem makes no claim about the existence nor absence of singularities—a unique maximal development exists, regardless of what the cause may be which makes it inextendible.

The last point we emphasise here is very different in flavour to the above. Namely, we should consider the implications that using Zorn's lemma to prove the Choquet-Bruhat–Geroch theorem has on the axiomatic foundation of general relativity. Famously, Zorn's lemma is equivalent to the axiom of choice—a controversial axiom. To understand why it is controversial, we should remind ourselves that axioms lay at the most foundational level of mathematics. Rather than being results that we prove, they are statements we *assume* to be true and build all of mathematics—and by that, theoretical physics—on top of. Naturally, such a choice of axioms should have a few properties: they should be consistent with each other, i.e., produce no contradictions, and they should allow us to prove as many theorems as possible. Further, an axiomatic system should be *minimal*, that is, contain only as many axioms as strictly required—we are interested in *proving* that things are true, not in assuming that they are.

One reason why the axiom of choice is controversial is that many important results in mathematics can be proven without its invocation. This opens up the question of whether to even accept it as an axiom, as in many contexts, it seems unnecessary. If, however, the axiom of choice does come up in proofs—such as those of Proposition 5.2 and Proposition 5.3—we should consider if it is possible to prove the same result without it. Informally speaking, the axiom of choice guarantees the existence of objects without having to construct them. Ridding a proof of this axiom can hence sometimes be achieved by explicitly constructing the desired object—such as the maximal element of a partially ordered set, rather than employing Zorn's lemma.

In the case of the Choquet-Bruhat–Geroch theorem, an alternative, constructive proof of the same result was given by Sbierski in [2] (2015). Though it might at first seem unnecessary to re-prove an already proven result, this more modern proof has profound consequences. Specifically, it shows that the axiom of choice can be removed from the axiomatic foundation of general relativity, hence reducing its size.

Regardless of how it is proven, though, the Choquet-Bruhat–Geroch theorem is a central result in mathematical relativity. It provides the rigorous mathematical groundwork for determinism within the theory and assures us of uniqueness of the global structure of spacetimes.

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